On Boundaries of Evidence

— Boundary-State Calculus for Typed Observation, Admissible Transfer, and Falsifiable Persistence

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Abstract

Boundary-State Calculus (BSC) is developed here as a typed audit calculus for finite open systems. Its purpose is not to supply a new ontology, but to decide when a claim remains warranted after boundary observation, quotienting, instrument change, transport, coarse-graining, dynamics, reconstruction, composition, and perturbation. We transcribe the hash-identified internal BSC system objects, subject to the public source-access limitation, and make explicit their measurable repair. We add a typed morphism record containing state and observable transport, stochastic simulation, equation residual, naturality defect, directed statistical deficiency, physical completion, and a certificate. Composition theorems show how defects and deficiencies accumulate and when a persistence certificate succeeds, becomes inconclusive, or is refuted.

The calculus incorporates boundary trace and Dirichlet-to-Neumann response, operational quantum quotients, Blackwell–Le Cam comparison of experiments, Koopman residuals and ordered learnability limits, sheaf descent, a conditional structured-cospan schema, non-invertible defects, and rigorous scale limits. For certified finite families it adds normalized scale profiles, normalization-covariant logarithmic rates, exceptional-set and slice singularities, contour-certified analytic transfer, and exact-decision bounds. It also defines a claim-relative simulation-evidence profile with typed loss coordinates, hard gates, joint statistical coverage, factored evidence identity, compatibility reserves, and finite-horizon surrogate propagation. For fixed compatible classical/quantum preparation-to-report pipelines, an operational-channel core propagates implemented reachable-set defects through Markov kernels, CPTP channels, measurements, and reporting. It proves downstream non-resurrection, a driven open-system energy identity, conditional Bernoulli information and zero-error bounds, and a single-identity relation alignment criterion. These are report-level consequences, not a unified microscopic theory, and they do not close the open full quantum BSC morphism. It separates topological charge from physical charge by requiring an explicit bridge derived from a gauge structure, action, current, anomaly, boundary condition, or measurement protocol. A claim-local mathematical verdict, five readiness coordinates, and a proof-dependency graph prevent local success from promoting unrelated claims. Ten exact fixtures expose both admissible transfers and hard obstructions, including deterministic receipts for a permanent regression counterexample and stable-host surrogate dependence. The resulting formalism treats persistence as an operational organizing principle: a persistent object is a recursively maintained finite boundary-state, and persistent identity is operational equivalence maintained under repeated admissible folds across declared scales. This principle is defined and made falsifiable; it is not asserted as a universal law.

Contents

1 Scope and non-claims

BSC concerns a finite open system only after "finite," "open," and "system" have been operationally fixed. Finiteness may mean bounded spatial domain, finite observation horizon, finite resources, or finite state cardinality; these meanings are not interchangeable. Openness means that a declared interface permits exchange with an exterior, not that all exterior degrees of freedom are known.

Definition 1.1 (Audit calculus)**.** BSC is the collection of typed objects, admissibility conditions, residuals, loss measures, dependency rules, and certificates used to assess a claim about a finite open system under observation, representation, transfer, scale, dynamics, reconstruction, composition, and perturbation.

This definition is deliberately narrower than several interpretations that have accumulated around boundary language. BSC is not a fundamental ontology, a unified field theory, a proof of holography or quantum gravity, a theory of consciousness, or a license to identify systems by visual resemblance. A boundary may be geometric, causal, administrative, statistical, or instrumental; the type must be stated. The words "topological," "recursive," and "quantum" carry no evidential force without their mathematical objects and physical bridges.

Convention 1.2 (Local promotion)**.** Evidence promotes only the claim node tested and those descendants whose declared dependencies are satisfied. It does not promote the framework, a neighboring claim, or an application by association.

Convention 1.3 (Source discipline)**.** An internal derivation, peer-reviewed result, preprint, numerical execution, and analogy are distinct source states. A citation supplies provenance; it does not supply a missing map or proof.

The cited core corpus contains both durable mathematics and explicit later repairs. The relevant lineage is transcribed from the hash-identified internal source set [Boundary-State Research Program, 2026, Boundary-State Calculus, a,b, Tree, 2026a], which is not independently inspectable in this public release. This paper records that transcription and its repairs, and excludes civic or metaphysical material from mathematical inference. The lineage claims are therefore source-local, not an independently replayable textual comparison. No empirical result is reported from the corpus.

2 Nomenclature, types, and front ledger

2.1 Basic types

Unless stated otherwise, measurable spaces are standard Borel. A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ on Y measurably in x. Kernel composition is

$$(L \odot K)(x, A) = \int_Y L(y, A)\, K(x, dy).$$

A measurable map $f : X \to Y$ is identified with its Dirac kernel δ_f. We write $B_b(X)$ for the bounded Borel measurable real- or complex-valued functions on X. Norms, metrics, and units are declared locally; quantities of unlike units are never added merely by choosing coefficients.

Table 1: Core nomenclature and types.

Symbol	Type	Role
Ω_ℓ	bounded domain or finite carrier	physical or abstract interior at scale ℓ; its meaning and units are declared
$\partial_\varepsilon\Omega_\ell$	measurable subdomain, or $\partial\Omega_\ell$ with trace regularity	finite sensor layer; never silently identified with an ideal boundary trace
$X_\ell, U_\ell, Y_\ell, C_\ell$	standard Borel spaces	raw states, controls, outputs, and contexts/calibrations
$\sim_\ell$	measurable equivalence relation on X_ℓ	observational, gauge, or representational equivalence; quotient $\bar{X}_\ell = X_\ell/\sim_\ell$ used only when standard Borel
$\mathcal{F}_{\ell,t}$	increasing sigma-algebras	legal experiment information through index t; pre-outcome information is $\mathcal{F}_{\ell,t-}$
$K^X_{\ell,t}, H^X_{\ell,t}$	$X_\ell \times U_\ell \times C_\ell \rightsquigarrow X_\ell, Y_\ell$	raw dynamics and observation kernels before quotient descent
$K^{\mathrm{dyn}}_{\ell,t}$	$\bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow \bar{X}_\ell$	descended local dynamics after lumpability; superscript prevents collision with interscale post-processing
$H^{\mathrm{obs}}_{\ell,t}$	$\bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow Y_\ell$	descended instrument channel after quotient invariance
G_ℓ	measurable loss/metric structure	comparison geometry with units and domain stated
$V_\ell \subseteq \bar{X}_\ell$	measurable admissible region	viability, safety, or model-validity domain
R_ℓ	declared family of typed kernels/maps	inherited class of scale, quotient, representation, or reconstruction maps; each use is namespaced below
B_ℓ	typed ledger	boundary flux, entropy, information, calibration, degeneracy, and provenance entries
$\mathcal{O}_\ell$	vector space, C^*-algebra, or measurable function class	declared accessible observables
$\tau_\partial, \Lambda_\Omega, E_\partial, e_\partial$	trace, response, reconstruction relation, selected extension	distinct boundary operators; their domains are not interchangeable
$\mathfrak{M}_{\ell m}$	typed morphism record	state/observable transfer, loss, defects, completion, and certificate
ORE	operational report envelope	fixed typed preparation, drive, measurement, report-law, decision, and certificate chain
$\delta(\mathsf{E}, \mathsf{F})$	number in $[0, 1]$	directed deficiency: ability of experiment E to simulate F
Cert	finite structured record	assumptions, tolerances, hashes, proof/execution receipts, sources, obligations, and status vector

2.2 Verdict and independent readiness axes

Definition 2.1 (Status vector)**.** Every claim c carries

$$\mathsf{Stat}(c) = (v_{\text{math}}, s_{\text{math}}, s_{\text{emp}}, s_{\text{comp}}, s_{\text{src}}, s_{\text{tr}}).$$

The claim-local mathematical verdict is

$$v_{\text{math}} \in \{\text{ill-posed}, \text{open}, \text{true}, \text{false}\}.$$

These four verdicts are not ordered. Only evidence about c itself may change $v_{\text{math}}(c)$; in particular, a false premise does not make a dependent claim false. The remaining coordinates are readiness chains:

$$\begin{aligned}
s_{\text{math}} &\in \{\text{none}, \text{conjectural}, \text{conditional}, \text{proved}\},\\
s_{\text{emp}} &\in \{\text{contradicted}, \text{untested}, \text{single study}, \text{replicated}\},\\
s_{\text{comp}} &\in \{\text{failed}, \text{unexecuted}, \text{executed}, \text{exact receipt}\},\\
s_{\text{src}} &\in \{\text{unchecked}, \text{internal}, \text{present proof}, \text{verified preprint}, \text{verified publication}\},\\
s_{\text{tr}} &\in \{\text{blocked}, \text{local only}, \text{bounded}, \text{certified}\}.
\end{aligned}$$

Each readiness set is ordered from least to greatest support and is denoted L_j; its meet is $\wedge_j$. An axis may instead be marked not applicable, in which case no edge may depend on it. Verdicts are never met, and readiness coordinates are never averaged. When monotonicity of a map from Stat is invoked, the verdict coordinate has the discrete order and the readiness coordinates have their displayed product order.

The ordering is evidential, not rhetorical. For example, a proved theorem can remain empirically untested; an exact execution can implement a claim whose encoding is ill-posed; a peer-reviewed source can be irrelevant to the asserted bridge.

2.3 Dependency graph

A claim ledger is a directed acyclic graph $D = (\mathcal{C}, E)$. An edge $e : a \xrightarrow{J_e} b$ states that claim b depends on claim a on the readiness set $J_e \subseteq \{\text{math}, \text{emp}, \text{comp}, \text{src}, \text{tr}\}$. For every named coordinate it also declares a monotone cap

$$\kappa_{e,j} : \mathsf{Stat}(a) \longrightarrow L_j.$$

Readiness propagates only through these caps. A false indispensable premise may make $\kappa_{e,j}$ the bottom element and thereby block transfer, but it does not assign the verdict false to b.

Definition 2.2 (Readiness-cap operator)**.** Let $r_j^0(b) \in L_j$ be claim b's readiness before dependency propagation. In a topological ordering of the claim DAG, define

$$r_j^*(b) = r_j^0(b) \wedge_j \bigwedge_{\substack{e:a\to b\\ j\in J_e}} \kappa_{e,j}(\mathsf{Stat}^*(a)),$$

where the empty meet is the top of L_j. An independent certificate is represented by changing the edge set or its cap, not by ignoring a stated dependency. Here $\mathsf{Stat}^*(a)$ combines a's unchanged verdict with its recursively capped readiness coordinates. The verdict is unchanged by this operator. Missing type data makes the claim-local verdict ill-posed and mathematical support none; missing execution lowers only computational readiness; a failed transfer lowers only transfer readiness unless it also supplies claim-local mathematical evidence.

Table 2: Concise claim-status ledger at entry. Every cell uses exactly one status from definition 2.1, or N/A when that axis is inapplicable.

Claim	Verdict	Math support	Empirical	Computation	Source	Transfer
Kernel quotient descent	true	proved	N/A	N/A	present proof	certified
Exact finite-label decoding criterion	true	proved	N/A	unexecuted	present proof	certified
Morphism defect accumulation	true	proved	N/A	unexecuted	present proof	bounded
Normalized scale-profile calculus	true	proved	N/A	unexecuted	present proof	certified
Simulation evidence profile and deployment admission	true	proved	N/A	unexecuted	present proof	bounded
Fixed-interface operational report-channel bound	true	proved	N/A	unexecuted	present proof	bounded
Channel form alone determines $\alpha^{-1} \approx 137$	false	proved	N/A	unexecuted	present proof	blocked
Zeta tail, rate, slice, contour, and local-root results	true	proved	N/A	unexecuted	present proof	bounded
Universal persistence principle	open	conjectural	untested	unexecuted	internal	local only
Universal boundary reconstruction	false	proved	N/A	unexecuted	present proof	blocked
Specified Koopman impossibility classes	true	proved	N/A	N/A	verified publication	local only
Winding determines electric charge	false	proved	untested	unexecuted	present proof	blocked
Klingman toroidal model yields QCD	ill-posed	proved	untested	unexecuted	verified publication	blocked
Fixtures F1–F7	true	proved	N/A	unexecuted	present proof	local only
Fixture F8 universal claim	false	proved	N/A	exact receipt	present proof	local only
Fixture F9 finite zeta identity	true	proved	N/A	unexecuted	present proof	bounded
Fixture F10 host-relative surrogate tolerance	true	proved	N/A	exact receipt	present proof	local only

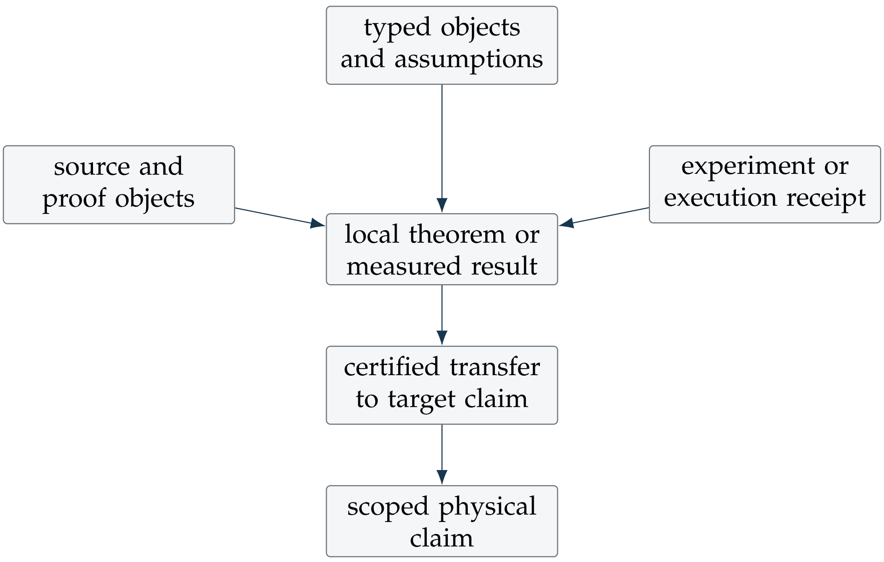

Figure 1: Minimal dependency architecture. Failure of an upstream requirement caps only the readiness of descendants reached by the edge's declared axes.

3 Finite boundary-state systems

3.1 Transcribed lineage objects

The hash-identified internal source gives the following original system object, transcribed here but not independently text-comparable from the public release:

$$S_\ell(t) = (\Omega_\ell, \partial_\varepsilon\Omega_\ell, X_\ell(t)/\sim_\ell, U_\ell(t), Y_\ell(t), H_\ell, G_\ell, V_\ell, R_\ell, K_\ell, B_\ell, \mathrm{rent}_\ell)\,. \tag{1}$$

It records a finite domain, observed boundary layer, quotient state, controls, outputs, observation, comparison geometry, viability, transformation class, local dynamics, boundary ledger, and an admission penalty [Boundary-State Research Program, 2026]. It is historically authoritative but under-typed: the context used by H_ℓ and the legal filtration are absent, R_ℓ is overloaded, and rent_ℓ has no fixed codomain.

The internal corpus then gives the following measurable repair, likewise transcribed with the same source-status limitation:

$$\widehat{S}_\ell = (\Omega_\ell, \partial_\varepsilon\Omega_\ell, X_\ell, \sim_\ell, U_\ell, Y_\ell, C_\ell, \mathcal{F}_\ell, K_\ell, H_\ell, G_\ell, V_\ell, R_\ell, B_\ell)\,. \tag{2}$$

The recorded lineage says that Volume III works with (2) while retaining the application layers of the first object [Boundary-State Calculus, a,b]. Subject to the inability to inspect those internal sources in this public release, we take $\widehat{S}_\ell$, not a newly invented tuple, as the current core.

Definition 3.1 (Typed interpretation of $\widehat{S}_\ell$)**.** Ω_ℓ is a declared bounded domain or finite carrier; $\partial_\varepsilon\Omega_\ell$ is a measurable layer, with ε carrying the domain's length or resolution unit. $X_\ell, U_\ell, Y_\ell, C_\ell$ are standard Borel spaces. In particular, C_ℓ is a structured physical-configuration space containing the instrument, observer or reference-frame state, controller, clock, and calibration state needed by the experiment; omitting one of these factors is a declared quotient, not a claim that the factor is nonphysical. $\sim_\ell$ is a measurable exact equivalence and $\bar{X}_\ell = X_\ell/\sim_\ell$ is required to be standard Borel whenever ordinary quotient kernels are used. An instantiated experiment declares a probability space $(\Xi_\ell, \mathcal{A}_\ell, \mathsf{P}_\ell)$ and an increasing filtration $\mathcal{F}_{\ell,t} \subseteq \mathcal{A}_\ell$, with $\mathcal{F}_{\ell,t-} = \sigma(\bigcup_{s<t} \mathcal{F}_{\ell,s})$ when the time order admits this notation.

$$K^X_{\ell,t} : X_\ell \times U_\ell \times C_\ell \rightsquigarrow X_\ell, \qquad H^X_{\ell,t} : X_\ell \times U_\ell \times C_\ell \rightsquigarrow Y_\ell$$

are the raw experiment kernels. Write $q_\ell : X_\ell \to \bar{X}_\ell$. The observation kernel must be invariant,

$$x \sim_\ell x' \Longrightarrow H^X_{\ell,t}(\cdot \mid x, u, c) = H^X_{\ell,t}(\cdot \mid x', u, c),$$

and the dynamics must be lumpable,

$$x \sim_\ell x' \Longrightarrow (q_\ell)_\# K^X_{\ell,t}(\cdot \mid x, u, c) = (q_\ell)_\# K^X_{\ell,t}(\cdot \mid x', u, c).$$

Only then do quotient descent and the final sigma-algebra define the tuple's operational kernels

$$K^{\mathrm{dyn}}_{\ell,t} : \bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow \bar{X}_\ell, \qquad H^{\mathrm{obs}}_{\ell,t} : \bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow Y_\ell.$$

G_ℓ is a declared metric, divergence, or measurable positive semidefinite bilinear-form field; $V_\ell \subseteq \bar{X}_\ell$ is measurable. R_ℓ is the inherited registry of typed scale, quotient, representation, and reconstruction maps, never a single polymorphic arrow. B_ℓ is a schema of ledger entries, each with a value, unit, method, uncertainty, time, and provenance. For legal preparation and feedback, let $(\mathsf{H}_{\ell,t}, \mathscr{H}_{\ell,t})$ be a standard Borel history space and let

$$\mathbf{H}_{\ell,t} : (\Xi_\ell, \mathcal{F}_{\ell,t-}) \longrightarrow (\mathsf{H}_{\ell,t}, \mathscr{H}_{\ell,t})$$

be the measurable history random variable. A history policy is a kernel

$$\pi_{\ell,t} : (\mathsf{H}_{\ell,t}, \mathscr{H}_{\ell,t}) \rightsquigarrow U_\ell \times C_\ell.$$

The implemented random kernel $\pi_{\ell,t}(\mathbf{H}_{\ell,t}, \cdot)$ is $\mathcal{F}_{\ell,t-}$-measurable and hence adapted. If a preparation does not factor through $\mathbf{H}_{\ell,t}$, it must instead be declared as an $\mathcal{F}_{\ell,t-}$-measurable random kernel on Ξ_ℓ. A state-Markov policy is separately denoted $\bar{\pi}_{\ell,t} : \bar{X}_\ell \rightsquigarrow U_\ell \times C_\ell$. Such a policy, or a sufficient history augmentation of $\bar{X}_\ell$, is required before the joint kernels below induce a state-marginal fold.

Definition 3.2 (System certificate companion)**.** Without changing (2), associate

$$\mathscr{C}_\ell = (D_\ell^{\text{rec}}, \mathsf{Prov}_\ell, \mathsf{Cert}_\ell),$$

where D_ℓ^{rec} contains reconstruction relations and their ambiguity classes; Prov_ℓ contains source identifiers, calibration lineage, data and code hashes; and Cert_ℓ contains assumptions, boundary conditions, tolerances, proof or execution objects, unresolved obligations, and Stat. The pair $(\widehat{S}_\ell, \mathscr{C}_\ell)$ is an instantiated certified system, not a replacement system tuple.

Definition 3.3 (Experiment bundle)**.** The retained experiment object is

$$\mathcal{E} = (\mathcal{S}, \mathcal{T}, \mathcal{I}, \mathcal{M}_0, \mathcal{O}, \mathcal{N}, \mathcal{L}, \mathcal{A}, \mathcal{D}).$$

Here S is (2); the repaired target is

$$\mathcal{T}_{\text{src}} = (Y, H, \tau, \mathcal{F}_t^{\text{legal}}),$$

where the source explicitly calls H the horizon but leaves τ undefined. We set $h := H$, choose a declared finite sampling set $J_{t,h} \subset [t, t+h]$, and introduce a measurable target functional

$$q : (Y^{J_{t,h}}, \mathcal{B}(Y)^{\otimes J_{t,h}}) \to Z_q.$$

We then use the explicit repair

$$\mathcal{T}^{\text{rep}} = (Y, h, q, \mathcal{F}_t^{\text{legal}}).$$

The observation channel remains in S and is not duplicated in the target. $\mathcal{I}$ is the intervention class; $\mathcal{M}_0$ a baseline; $\mathcal{O}$ candidate operators; $\mathcal{N}$ matched nulls; $\mathcal{L}$ a scoring rule; $\mathcal{A}$ an admission rule; and

$$\mathcal{D} : \text{Claims} \times \text{Evidence} \to \{\text{admit}, \text{sandbox}, \text{watch}, \text{demote}, \text{retire}\}.$$

The name "demotion map" is retained, although its codomain also includes admission.

3.2 Typed folds

The original fold $\Pi_{V_m}^{G_m} \circ R_{\ell m} \circ H_\ell \circ K_\ell$ mixes state dynamics, observation, inference, and state transport. Its architecture is useful but its displayed composite is not type-correct.

Definition 3.4 (Lifted observation fold)**.** Put $\mathsf{Z}_\ell^{\text{sc}} = \bar{X}_\ell \times U_\ell \times C_\ell$. Define

$$\widetilde{K}_\ell((x,u,c), d(x',u',c')) = K_\ell^{\text{dyn}}(dx' \mid x,u,c)\delta_u(du')\delta_c(dc'),$$
$$\widetilde{H}_\ell((x,u,c), d(y,c')) = H_\ell^{\text{obs}}(dy \mid x,u,c)\delta_c(dc').$$

If $R_{\ell m}^Y : Y_\ell \times C_\ell \rightsquigarrow \bar{X}_m$ is an inference kernel and $\Pi_{V_m}^{G_m} : \bar{X}_m \rightsquigarrow V_m$ a declared measurable repair or projection kernel, then

$$F_{\ell m}^{\text{obs}} = \Pi_{V_m}^{G_m} \odot R_{\ell m}^Y \odot \widetilde{H}_\ell \odot \widetilde{K}_\ell : \mathsf{Z}_\ell^{\text{sc}} \rightsquigarrow V_m$$

is a typed observation-to-state fold. A direct state coarse-graining $R_{\ell m}^X : \bar{X}_\ell \rightsquigarrow \bar{X}_m$ defines a separate route

$$F_{\ell m}^X = \Pi_{V_m}^{G_m} \odot R_{\ell m}^X \odot \delta_{\pi_X} \odot \widetilde{K}_\ell, \qquad \pi_X : \mathsf{Z}_\ell^{\text{sc}} \to \bar{X}_\ell.$$

Conflating these routes hides whether information was discarded by the instrument, the inference, the quotient, or the scale map.

4 Observation events, filtration, and leakage control

Definition 4.1 (Observation atom)**.** The retained observation atom is

$$a_t = (\ell, t, \Omega, \partial_\varepsilon\Omega, x_t, u_t, c_t, H_t, y_t, \Sigma_{y,t}, \mathcal{F}_t).$$

It is legal for prediction horizon $h > 0$ when $x_t \in \bar{X}_\ell$, $y_t \in Y_\ell$, and $H_t = H^{\mathrm{obs}}_{\ell,t}$. When $Y_\ell \subseteq \mathbb{R}^d$, $\Sigma_{y,t} \in \mathbb{R}^{d\times d}$ is positive semidefinite and has declared output units squared; for a general output space it is replaced by a typed uncertainty object. The displayed atom is a realization of jointly defined random objects $\mathbf{X}_{\ell,t}, \mathbf{U}_{\ell,t}, \mathbf{C}_{\ell,t}, \mathbf{Y}_{\ell,t}, \mathbf{\Sigma}_{\ell,t}$ on $(\Xi_\ell, \mathcal{A}_\ell, \mathsf{P}_\ell)$. Every design, preparation, and statistic used before observing $\mathbf{Y}_{\ell,t}$, including every predictor of $\mathbf{Y}_{\ell,t+h}$, is measurable with respect to the declared pre-outcome sigma-algebra $\mathcal{F}_{\ell,t-}$.

Definition 4.2 (Boundary event)**.** For $r \in \partial\Omega$,

$$e_t = (r, t, J_k(r,t), X_k(r,t), \nu(r), \tau(r))$$

records a flux/current component J_k, boundary field X_k, outward normal ν, and orientation or time label τ. Their tensor types and units must be fixed by the governing model before any conservation checksum is formed.

Assumption 4.3 (Pre-outcome freeze)**.** For an outcome at time t, the design, preprocessing graph, dictionary, candidate operators, null library, thresholds, and seed policy are $\mathcal{F}_{\ell,t-}$-measurable random objects on Ξ_ℓ. Holdout outcomes and post-outcome edits are not $\mathcal{F}_{\ell,t-}$-measurable.

Definition 4.4 (Leakage defect)**.** Let $\mathbf{Z} : \Xi_\ell \to \mathcal{Z}$ denote the complete fitted artifact and $\mathbf{Y}^{\mathrm{hold}} : \Xi_\ell \to \mathcal{Y}_{\mathrm{hold}}$ the held-out outcome. A protocol has structural leakage if $\mathbf{Z}$ is not $\mathcal{F}_{\ell,t-}$-measurable. A probabilistic diagnostic under P_ℓ is

$$\lambda_{\mathrm{leak}} = I_{\mathsf{P}_\ell}(\mathbf{Z}; \mathbf{Y}^{\mathrm{hold}} \mid \mathcal{F}_{\ell,t-}),$$

when the conditional mutual information exists. The diagnostic being zero does not prove correct measurability; positive value refutes isolation.

The earlier corpus condition that two sigma-algebras have empty intersection is impossible because both contain the empty set and sample space. The repair is conditional isolation plus a logged access history. If assumption 4.3 fails, computational or empirical promotion based on that holdout is demoted regardless of predictive score.

Proposition 4.5 (Predictable sequential admission)**.** *Let $A_t \in \mathcal{A}_\ell$ be false-admission events on Ξ_ℓ, and let $\alpha_t : \Xi_\ell \to [0,1]$ be $\mathcal{F}_{\ell,t-1}$-measurable. If $\mathsf{P}_\ell(A_t \mid \mathcal{F}_{\ell,t-1}) \le \alpha_t$ almost surely and $\sum_t \alpha_t \le \alpha$, then $\mathsf{P}_\ell(\bigcup_t A_t) \le \alpha$.*

Proof. Taking expectations gives $\mathsf{P}_\ell(A_t) \le \mathbb{E}_{\mathsf{P}_\ell}\alpha_t$. The union bound and Tonelli's theorem give $\mathsf{P}_\ell(\bigcup_t A_t) \le \sum_t \mathbb{E}_{\mathsf{P}_\ell}\alpha_t \le \mathbb{E}_{\mathsf{P}_\ell}\sum_t \alpha_t \le \alpha$. □

This result does not validate the null model, exchangeability, or the physical meaning of the score. Those are separate dependencies.

5 Boundary trace, response, and reconstruction

Let $\Omega \subset \mathbb{R}^n$ be bounded with boundary regularity stated locally. The following three operations are distinct.

Definition 5.1 (Trace and layer)**.** For a Banach function space $\mathcal{X}(\Omega)$ admitting a continuous trace,

$$\tau_\partial : \mathcal{X}(\Omega) \to \mathcal{X}_\partial(\partial\Omega).$$

For finite sensor resolution,

$$\partial_\varepsilon\Omega = \{x \in \Omega : \mathrm{dist}(x, \partial\Omega) \le \varepsilon\}, \qquad L_\varepsilon : \mathcal{X}(\Omega) \to \mathcal{Y}_\varepsilon$$

is a separate measurement map. No equality $L_\varepsilon = \tau_\partial$ is presumed.

Definition 5.2 (Boundary response)**.** Let $L_a u = 0$ be a declared elliptic boundary-value problem, with coefficient a in an admissible class. Let $\mathcal{B}_D$ and $\mathcal{B}_N$ be the declared Dirichlet and conormal-response spaces. In the standard weak uniformly elliptic setting on a bounded Lipschitz domain one may take $\mathcal{B}_D = H^{1/2}(\partial\Omega)$ and $\mathcal{B}_N = H^{-1/2}(\partial\Omega)$. If the weak solution u_f is well posed for every $f \in \mathcal{B}_D$, the Dirichlet-to-Neumann map is

$$\Lambda_{a,\Omega} : \mathcal{B}_D \to \mathcal{B}_N, \qquad f \longmapsto \partial_{\nu,a} u_f\big|_{\partial\Omega},$$

where $\partial_{\nu,a}$ is the model's conormal derivative. Thus $\Lambda_{a,\Omega}$ maps imposed boundary data to measured response; it is not the trace of an arbitrary interior state.

Definition 5.3 (Reconstruction relation)**.** For a data class $\mathcal{D}$ and admissible interior class $\mathcal{A}$,

$$E_\partial \subseteq \mathcal{D} \times (\mathcal{A}/\!\approx)$$

is a possibly set-valued reconstruction relation. The equivalence $\approx$ contains every gauge or boundary-fixing diffeomorphism ambiguity left invariant by the data. Identifiability means that each data item in the range has at most one equivalence class; stability is a separate bound

$$d_{\mathcal{A}/\approx}(a_1, a_2) \le \omega(d_{\mathcal{D}}(\Lambda_{a_1}, \Lambda_{a_2}))$$

on a stated subset, for a declared modulus ω.

Definition 5.4 (Selected extension)**.** For a declared subset $\mathcal{D}_0 \subseteq \mathcal{D}$, a selected extension is a measurable map

$$e_\partial : \mathcal{D}_0 \to \mathcal{A} \subseteq \mathcal{X}(\Omega).$$

Writing $r : \mathcal{A} \to \mathcal{A}/\!\approx$ for the quotient map, it must satisfy

$$(d, r(e_\partial(d))) \in E_\partial \quad \text{for every } d \in \mathcal{D}_0.$$

Existence and measurability are assumptions; linearity and boundedness are additional assumptions, never consequences of the relation. Any quantity evaluated on $e_\partial(d)$ must either be invariant under $\approx$ or record the representative-selection dependence.

Scoped isotropic uniqueness theorems are available in the dimensions and regularity classes of the cited works [Sylvester and Uhlmann, 1987, Astala and Päivärinta, 2006]. In the anisotropic manifold class of Lassas and Uhlmann [2001], recovery is modulo the specified boundary-fixing geometric equivalence. Stability is weaker and separately conditional [Alessandrini, 1988]. None of these results licenses the phrase "the boundary determines the interior" without the PDE, data class, regularity, quotient, boundary conditions, and stability theorem.

Proposition 5.5 (Trace-limited prediction)**.** *Let $F : \mathcal{X}(\Omega) \to Z$ be L_F-Lipschitz and let a selected extension $e_\partial L_\varepsilon x$ be defined. Then*

$$\|F(x) - F(e_\partial L_\varepsilon x)\|_Z \leq L_F \|x - e_\partial L_\varepsilon x\|_{\mathcal{X}}.$$

Proof. This is the defining Lipschitz inequality. It is an upper bound, not an identifiability theorem. □

If additive data noise η is passed through a bounded linear selected extension, $\|e_\partial \eta\| \leq \|e_\partial\| \, \|\eta\|$. The operator norm supplies an upper worst-case amplification, not a universal lower error bound.

6 Observable algebras and operational quotients

6.1 Classical experiment quotients

Definition 6.1 (Exact observation equivalence)**.** For legal contexts and controls,

$$x \sim_H x' \quad \Longleftrightarrow \quad H^X(\cdot \mid x, u, c) = H^X(\cdot \mid x', u, c) \quad \text{for all legal } (u, c).$$

A query $q : X \to Z$ descends when it is constant on these classes.

Theorem 6.2 (Kernel quotient descent)**.** *Let $q : X \to \bar{X} = X/\sim$ be a measurable quotient of standard Borel spaces, with the final sigma-algebra, and let $M : X \rightsquigarrow Y$ be a kernel constant on equivalence classes such that every coordinate $M(\,\cdot\,, A)$ is q-measurable. Then there is a unique kernel $\bar{M} : \bar{X} \rightsquigarrow Y$ satisfying $M = \bar{M} \odot \delta_q$.*

Proof. For A in a countable generator of $\mathcal{B}(Y)$, define $\bar{M}([x], A) = M(x, A)$. Constancy makes this well defined. Quotient measurability and a monotone-class argument extend it to every Borel A; countable additivity is inherited pointwise. Surjectivity of q gives uniqueness. □

If $\bar{X}$ is not standard Borel or stabilizer information matters, an ordinary quotient is inadequate; a measurable groupoid or invariant-algebra description is required.

Definition 6.3 (Confusability pseudometric)**.** Given a probability metric d_P,

$$d_E(x, x') = \sup_{(u,c) \text{ legal}} d_P\Big(H^{\text{obs}}(\cdot \mid x, u, c), H^{\text{obs}}(\cdot \mid x', u, c)\Big).$$

The set $d_E(x, x') \leq \varepsilon$ is an identified neighborhood or confusability graph relation. It is not generally transitive and therefore does not generally define a quotient.

Proposition 6.4 (Exact decision descent and stochastic separation)**.** *Let $\Sigma \subseteq X$ be measurable, let $R : X \to Y$ be a measurable deterministic report, and put $q_\Sigma = \mathbf{1}_\Sigma$. There is a measurable $d : Y \to \{0, 1\}$ with $q_\Sigma = d \circ R$ if and only if there is a measurable $B \subseteq Y$ such that*

$$\Sigma = R^{-1}(B).$$

In particular, a report fiber meeting both Σ and its complement blocks exact decision descent.

More generally, let Q be finite with its discrete sigma-algebra, let $q : X \to Q$ be measurable, and let $O : X \rightsquigarrow Y$ be a kernel on a standard Borel output space. An exact decoder $d : Y \to Q$, satisfying

$$O\big(x, d^{-1}(\{q(x)\})\big) = 1 \quad \textit{for every } x,$$

exists if and only if a measurable partition $(B_a)_{a \in Q}$ of Y satisfies $O(x, B_{q(x)}) = 1$ for every x. Hence laws associated with different labels must be mutually singular. For x_0, x_1 with $q(x_0) \neq q(x_1)$, put

$P_j = O(x_j, \cdot)$. *Under equal priors, every binary decoder has average, and therefore worst-case, error at least*

$$\frac{1 - d_{\mathrm{TV}}(P_0, P_1)}{2}, \qquad d_{\mathrm{TV}}(P_0, P_1) = \sup_B |P_0(B) - P_1(B)|.$$

Proof. For the deterministic statement, a factorization gives $B = d^{-1}(\{1\})$; conversely $d = \mathbf{1}_B$ gives the factorization. The fibers of an exact finite-label decoder form the required partition, and a declared partition defines the decoder. Distinct labels then concentrate on disjoint cells.

For the binary bound, if B is the decision region for label 0, the equal-prior error is

$$\frac{1}{2}\{P_0(B^c) + P_1(B)\} = \frac{1}{2}\{1 - (P_0(B) - P_1(B))\} \geq \frac{1 - d_{\mathrm{TV}}(P_0, P_1)}{2}.$$

The maximum of the two conditional errors is at least their average. □

Identical output laws therefore make exact classification impossible, while close laws require a quantitative testing statement. Repeated observations replace P_0, P_1 by the declared product or dependent joint laws and must record the sample count; repetition is not inherited for free.

6.2 Quantum operational quotients

Let $\mathcal{H}$ be finite-dimensional for the present definition, and let $\mathcal{A}_{\mathrm{acc}} \subseteq \mathcal{B}(\mathcal{H})$ be the declared operator system of accessible effects.

Definition 6.5 (Operational quantum equivalence)**.** For density operators ρ, σ,

$$\rho \sim_{\mathcal{A}_{\mathrm{acc}}} \sigma \quad \iff \quad \sup_{\substack{0 \leq E \leq I \\ E \in \mathcal{A}_{\mathrm{acc}}}} |\operatorname{Tr} E(\rho - \sigma)| = 0.$$

This is exact operational equivalence: equality of accessible probabilities, not equality of matrix representations. The pseudometric

$$d_{\mathcal{A}_{\mathrm{acc}}}(\rho, \sigma) = \sup_{\substack{0 \leq E \leq I \\ E \in \mathcal{A}_{\mathrm{acc}}}} |\operatorname{Tr} E(\rho - \sigma)|$$

defines ε-confusability by $d_{\mathcal{A}_{\mathrm{acc}}}(\rho, \sigma) \leq \varepsilon$. For $\varepsilon > 0$ confusability is generally nontransitive and is not called an equivalence.

A state channel $\Phi : \mathcal{T}(\mathcal{H}_\ell) \to \mathcal{T}(\mathcal{H}_m)$ travels with its Heisenberg pullback

$$\Phi^\sharp : \mathcal{B}(\mathcal{H}_m) \to \mathcal{B}(\mathcal{H}_\ell), \qquad \operatorname{Tr}[\Phi(\rho) A] = \operatorname{Tr}[\rho\, \Phi^\sharp(A)].$$

Quantum reference-frame changes may alter which subalgebras count as subsystems or accessible observables [Carette et al., 2025, Ahmad et al., 2022, Castro-Ruiz and Oreshkov, 2025]. Discarding is therefore relative to a declared algebra and factorization, not an observer-free deletion operation.

Definition 6.6 (Transform–discard defect)**.** Let $\Phi : \mathcal{T}(\mathcal{H}_R \otimes \mathcal{H}_S) \to \mathcal{T}(\mathcal{H}_{R'} \otimes \mathcal{H}_{S'})$ be a frame or description channel. Suppose a candidate reduced channel $\bar{\Phi} : \mathcal{T}(\mathcal{H}_S) \to \mathcal{T}(\mathcal{H}_{S'})$ is declared. On accessible effects $\mathcal{A}_{S'}$,

$$\eta_{\mathrm{td}}(\rho) = \sup_{\substack{0 \leq A \leq I \\ A \in \mathcal{A}_{S'}}} \left| \operatorname{Tr} A \, \operatorname{Tr}_{R'} \Phi(\rho) - \operatorname{Tr} A \, \bar{\Phi}(\operatorname{Tr}_R \rho) \right|.$$

Transform and discard commute operationally on a class $\mathcal{Q}$ when a single $\bar{\Phi}$ makes

$$\sup_{\rho \in \mathcal{Q}} \eta_{\mathrm{td}}(\rho) = 0.$$

Proposition 6.7 (Paired transport preserves probabilities)**.** *For every state ρ and target effect A,* $\mathrm{Tr}[\Phi(\rho)A] = \mathrm{Tr}[\rho\Phi^\sharp(A)]$. *Consequently a representation change is operationally harmless only for probabilities tested with the paired observable transport.*

Proof. This is the definition of the Banach adjoint $\Phi^\sharp$. □

7 Admissible morphisms and composition

The hash-identified internal source is reported to contain the deterministic morphism

$$\mu_{a\to b} = (R_{a\to b}, R_Y, \varepsilon_R, \tau_R, q_{a\to b})$$

which is transcribed here as pairing state and output maps with an error field, an otherwise untyped τ_R, and quotient compatibility [Boundary-State Research Program, 2026]. The internal source is not independently inspectable in this public release. In particular, τ_R is not silently promoted here to a certificate. The printed two-step error rule in that source is repaired to the type-correct downstream Lipschitz bound

$$\varepsilon_{ac} \le \varepsilon_{bc} + L_{bc}\varepsilon_{ab}.$$

The later budgeted record is reported in the supplied lineage and transcribed as

$$\mu = (R_X, R_Y, R_U, R_C, \alpha_H, \alpha_K, q_{ab}, b_\mu),$$

where q_{ab} records quotient compatibility and b_μ a distortion budget [Boundary-State Calculus, a]. Under explicit typing, R_X embeds in $T_{\ell m}$, R_Y in $K_{\ell m}$, R_U, R_C in the completion, $\alpha_H, \alpha_K, b_\mu$ in the defect/tolerance fields, and q_{ab} in the information-loss and admissibility fields. The older $R_{a\to b}, R_Y$ embed similarly; $\varepsilon_R, \tau_R, q_{a\to b}$ are preserved in the certificate until their types are discharged. The following record extends both; it does not replace the system object.

Definition 7.1 (Typed BSC morphism)**.** A morphism from certified system ℓ to certified system m is

$$\mathfrak{M}_{\ell\to m} = \left(T_{\ell m}, T^\sharp_{\ell m}, K_{\ell m}, R_{\ell m}, \Theta_{\ell m}, \delta_{\ell m}, C_{\ell m}, \mathsf{Cert}_{\ell m}\right), \tag{3}$$

with a variant tag $\mathsf{v} \in \{\mathrm{det}, \mathrm{stoch}, \mathrm{quant}, \mathrm{corr}, \mathrm{defect}\}$. For the stochastic variant:

1. $T_{\ell m} : D_{\ell m} \rightsquigarrow \bar{X}_m$ is a Markov kernel on a declared measurable domain $D_{\ell m} \subseteq \bar{X}_\ell$.
2. $$T^\sharp_{\ell m} : B_b(\bar{X}_m) \to B_b(D_{\ell m}), \qquad T^\sharp_{\ell m} g(x) = \int g(x')T_{\ell m}(x, dx')$$
   is positive and unital. It need not be multiplicative. In this bounded stochastic variant, $\mathcal{O}_i \subseteq B_b(\bar{X}_i)$, and the declared observable families must satisfy
   $$T^\sharp_{\ell m}(\mathcal{O}_m) \subseteq \mathcal{O}_\ell|_{D_{\ell m}};$$
   otherwise the state transfer does not transport the observables claimed. Unbounded observables require a separately declared invariant integrability domain.
3. $K_{\ell m} : Y_\ell \rightsquigarrow Y_m$ is an observation post-processing or simulation channel.
4. Given equation operators $\mathcal{E}_i : \mathsf{Dom}(\mathcal{E}_i) \to Z_i$ and an equation-space map $S_{\ell m} : Z_\ell \to Z_m$, the residual record $R_{\ell m}$ contains the target-equation residual, its Banach space, norm, units, evaluation class, and bound. In the deterministic variant,
   $$R_{\ell m}(x) = \mathcal{E}_m(T_{\ell m}x) - S_{\ell m}\mathcal{E}_\ell(x) \in Z_m.$$
   For stochastic or nonlinear equations, the certificate must specify the law-level weak residual; no canonical expectation is presumed.

5. Put $\mathsf{Z}_i^{\mathrm{sc}} = \bar{X}_i \times U_i \times C_i$. The completion $C_{\ell m}$ contains control and context transports and a joint lift
$$\widehat{T}_{\ell m} : \widehat{D}_{\ell m} \rightsquigarrow \mathsf{Z}_m^{\mathrm{sc}}, \qquad \widehat{D}_{\ell m} \subseteq \mathsf{Z}_\ell^{\mathrm{sc}},$$
whose state marginal agrees with $T_{\ell m}$ under the declared source policy. It also contains the physical carrier, controller, observer, reference frame, instrument, clock map, resource interface, and boundary conditions.
6. On the completed common domain,
$$\Theta_{\ell m} = \sup_{z \in \widehat{D}_{\ell m}} d_m\Big(H_m^{\mathrm{obs}} \odot \widehat{T}_{\ell m}(z), K_{\ell m} \odot H_\ell^{\mathrm{obs}}(z)\Big)$$
is a naturality or commuting-square defect in a declared probability metric.
7. For experiments $\mathsf{E}_i = \{P_{i,\theta} : \theta \in \Theta\}$,
$$\delta_{\ell m} = \delta(\mathsf{E}_\ell, \mathsf{E}_m) = \inf_{K: Y_\ell \rightsquigarrow Y_m} \sup_{\theta \in \Theta} \|KP_{\ell,\theta} - P_{m,\theta}\|_{\mathrm{TV}}.$$
The total-variation convention is $\|P - Q\|_{\mathrm{TV}} = \frac{1}{2}\|p - q\|_1$ in the finite case. For the particular implemented post-processor in the record, define
$$e_{\ell m}(K_{\ell m}) = \sup_{\theta \in \Theta} \|K_{\ell m} P_{\ell,\theta} - P_{m,\theta}\|_{\mathrm{TV}} \geq \delta_{\ell m}.$$
The certificate says whether this channel attains the infimum or is only a declared near-minimizer.
8. $\mathsf{Cert}_{\ell m}$ is the record specified in definition 3.2, augmented by variant, domains, tolerances, information destroyed, residual and defect evaluations, implementation evidence, and status vector.

A partial deterministic map uses the same record with explicit domain and failure output. A correspondence $T \subseteq X_\ell \times X_m$ requires a selection rule, multiplicity semantics, or probability kernel before observable pullback is single-valued. A quantum channel uses trace-class states and its completely positive unital adjoint. Defect and fusion variants are treated in section 12. These variants do not share a single unqualified composition theorem.

Definition 7.2 (Claim-relative admissibility)**.** A claim c supplies a specification
$$\mathcal{A}_c = (J_c, (\tau_{c,j})_{j \in J_c}, \mathcal{G}_c, \mathrm{Req}_c).$$
Here J_c names the coordinates of the loss vector $\mathcal{L}_{\ell m}$ in definition 8.1 that the claim uses, $\tau_{c,j}$ is the tolerance for coordinate j, $\mathcal{G}_c$ is a finite set of Boolean hard gates, and Req_c gives minimum readiness on the applicable status axes. The morphism is *well formed for c* only if its certificate answers all of the following:

1. the domain, codomain, variant, regularity, units, and boundary conditions of $T_{\ell m}$;
2. the target observable class and reverse-direction transport $T_{\ell m}^\sharp$;
3. the equivalence classes or information distinctions destroyed;
4. the target equations, equation-space map, residual norm, and tolerance;
5. the observation and other diagrams tested, including nonzero defects;
6. the physical carrier, controller, clock, observer, reference frame, and instrument that implement the transfer;
7. the assumptions, sources, proof objects, execution receipts, hashes, unresolved obligations, claim-local verdict, and readiness coordinates.

It is *evaluated for c* only if every $j \in J_c$ has a proved enclosure

$$I_{c,j} = [\underline{\ell}_{c,j}, \overline{\ell}_{c,j}] \quad \text{with} \quad \ell_{c,j}^{\text{actual}} \in I_{c,j},$$

and every hard gate has a certified truth value. It is *admissible for c* if and only if it is well formed and evaluated, $\overline{\ell}_{c,j} \le \tau_{c,j}$ for all $j \in J_c$, every gate in $\mathcal{G}_c$ is true, and its readiness meets Req_c.

A missing type makes the record not well formed; a missing enclosure or gate evaluation makes it unevaluated; a failed threshold, hard gate, or readiness requirement makes it inadmissible. If $\underline{\ell}_{c,j} > \tau_{c,j}$, the coordinate violates its tolerance. If $\underline{\ell}_{c,j} \le \tau_{c,j} < \overline{\ell}_{c,j}$, the enclosure is inconclusive: the certificate fails to establish admissibility, but the claim is not thereby refuted. None of these cases is represented as a small numerical error in a missing field.

7.1 Composition in the stochastic variant

Assumption 7.3 (Compatible completions)**.** For $\mathfrak{M}_{\ell m}$ and $\mathfrak{M}_{mn}$, the target interface, clock, resource contract, boundary conditions, control/context law, and carrier output of $C_{\ell m}$ match the source requirements of C_{mn}. The composite completion is their specified fiber product over the shared interface. On its eventual domain, the projection to source state lies in the state-composite domain, and its state marginal agrees with the composite state kernel under the declared composed policy. Correlated failure modes are retained.

Definition 7.4 (Stochastic composite)**.** Under assumption 7.3, define

$$K_{\ell n} = K_{mn} \odot K_{\ell m},$$

and give the state composite the maximal measurable domain

$$D_{\ell n} = \{x \in D_{\ell m} : T_{\ell m}(x, D_{mn}) = 1\}.$$

For $x \in D_{\ell n}$, set

$$T_{\ell n}(x, A) = \int_{D_{mn}} T_{mn}(y, A)\, T_{\ell m}(x, dy);$$

this is the partial-kernel composite denoted $T_{mn} \odot T_{\ell m}$. Define the restricted pullback

$$T_{\ell m}^{\sharp|D_{mn}} : B_b(D_{mn}) \longrightarrow B_b(D_{\ell n}), \qquad T_{\ell m}^{\sharp|D_{mn}} h(x) = \int_{D_{mn}} h(y)\, T_{\ell m}(x, dy).$$

The type-correct reverse composite is

$$T_{\ell n}^{\sharp} = T_{\ell m}^{\sharp|D_{mn}} \circ T_{mn}^{\sharp} : B_b(\bar{X}_n) \longrightarrow B_b(D_{\ell n}).$$

On state–control–context space put

$$\widehat{D}_{\ell n} = \{z \in \widehat{D}_{\ell m} : \widehat{T}_{\ell m}(z, \widehat{D}_{mn}) = 1\}.$$

The compatible completion defines, for $z \in \widehat{D}_{\ell n}$,

$$\widehat{T}_{\ell n} = \widehat{T}_{mn} \odot \widehat{T}_{\ell m}.$$

Its source-state projection lies in $D_{\ell n}$, and its state marginal is $T_{\ell n}$ under the declared composed policy, as required by assumption 7.3. The certificate is the conjunction of assumptions and obligations and the union of provenance DAGs. It preserves claim-local verdicts and computes descendant readiness through the declared caps of definition 2.2. A new hash identifies the composite record; it is not a new proof.

Proposition 7.5 (Partial pullback composition)**.** *The pullback in definition 7.4 is independent of any extension of $T^{\sharp}_{mn}g$ away from D_{mn}, and for every $g \in B_b(\bar{X}_n)$ it is the observable pullback induced by $T_{\ell n}$.*

Proof. For $x \in D_{\ell n}$, the measure $T_{\ell m}(x, \cdot)$ is supported on D_{mn}, so two bounded measurable extensions agreeing on D_{mn} have equal integrals. For $g \geq 0$, Tonelli's theorem gives

$$\int_{\bar{X}_n} g(w)\, T_{\ell n}(x, dw) = \int_{D_{mn}} \left[\int_{\bar{X}_n} g(w)\, T_{mn}(y, dw) \right] T_{\ell m}(x, dy).$$

This is $(T^{\sharp|D_{mn}}_{\ell m} T^{\sharp}_{mn} g)(x)$. Positive and negative parts, and then real and imaginary parts, extend the identity to every bounded real- or complex-valued g. $\square$

Theorem 7.6 (Observation-defect propagation)**.** *Let $d_i = \|\cdot\|_{\mathrm{TV}}$, and let $\eta(K_{mn}) \leq 1$ be the Dobrushin contraction coefficient of K_{mn}. For compatible stochastic morphisms,*

$$\Theta_{\ell n} \leq \Theta_{mn} + \eta(K_{mn})\Theta_{\ell m}.$$

For a Wasserstein metric, the same statement holds with the declared Lipschitz coefficient of K_{mn}, provided the required moments exist.

Proof. Apply the triangle inequality through the intermediate law $K_{mn} \odot H^{\mathrm{obs}}_m \odot \widehat{T}_{\ell m}$. The first difference is bounded by Θ_{mn} after averaging over $\widehat{T}_{\ell m}$; the second contracts by $\eta(K_{mn})$. The identity $\widehat{T}_{\ell n} = \widehat{T}_{mn} \odot \widehat{T}_{\ell m}$ supplies the other side of the composite square. Take the supremum. $\square$

Theorem 7.7 (Directed deficiency triangle)**.** *Suppose all three experiments have the same parameter set Θ, or have been pulled back to one by declared parameter maps. Then*

$$\delta(\mathsf{E}_\ell, \mathsf{E}_n) \leq \delta(\mathsf{E}_\ell, \mathsf{E}_m) + \delta(\mathsf{E}_m, \mathsf{E}_n).$$

Proof. Choose channels within ε of the two infima and compose them. Total variation contracts under Markov kernels; the triangle inequality gives the displayed bound plus 2ε. Let $\varepsilon \downarrow 0$. $\square$

Theorem 7.8 (Deterministic equation-residual propagation)**.** *For deterministic state maps with $T_{\ell n} = T_{mn}T_{\ell m}$ and $S_{\ell n} = S_{mn}S_{\ell m}$, suppose $S_{mn} : Z_m \to Z_n$ is bounded linear, the equation domains are compatible, and $T_{\ell m}(D_{\ell m}) \subseteq \mathsf{Dom}(R_{mn})$. Then*

$$R_{\ell n}(x) = R_{mn}(T_{\ell m}x) + S_{mn}R_{\ell m}(x).$$

Consequently,

$$\|R_{\ell n}\|_{\infty, D_{\ell m}} \leq \|R_{mn}\|_{\infty, T_{\ell m}(D_{\ell m})} + \|S_{mn}\|\, \|R_{\ell m}\|_{\infty, D_{\ell m}}.$$

Proof. Add and subtract $S_{mn}\mathcal{E}_m(T_{\ell m}x)$ in $\mathcal{E}_n(T_{mn}T_{\ell m}x) - S_{mn}S_{\ell m}\mathcal{E}_\ell(x)$. $\square$

The identity morphism has $T = \delta_{\mathrm{id}}$, $T^\sharp = \mathrm{id}$, $K = \delta_{\mathrm{id}}$, zero residual and defects, zero deficiency, identity completion, and a certificate containing the relevant system identities. Kernel associativity, reverse pullback composition, and the two fiber-product unit isomorphisms give the unit laws. If completion compatibility fails, the composite is undefined rather than high-error.

8 Defects, residuals, deficiency, and information loss

Residual, naturality defect, statistical deficiency, quotient loss, and viability failure live in different spaces. They are recorded as a vector, not collapsed into a mechanism score.

Definition 8.1 (Loss vector)**.** For a morphism, let $\mathcal{R}_{\ell m}$ be its typed residual-record space, let $\mathcal{Q}^{\mathrm{res}}_{\ell m}$ be a declared residual test class, and let

$$\mathsf{ev}^{\mathrm{res}}_{\ell m} : \mathcal{R}_{\ell m} \times \mathcal{Q}^{\mathrm{res}}_{\ell m} \to [0, \infty]$$

be a declared measurable scalar evaluator with recorded units. Let $\mathcal{Q}^{\mathrm{law}}_{\ell m}$ be a declared class of admissible input probability laws. Put

$$\mathcal{L}_{\ell m} = \left(\mathfrak{r}_{\ell m}, \Theta_{\ell m}, \delta_{\ell m}, e_{\ell m}(K_{\ell m}), \lambda_{\mathrm{quot}}, \alpha_{\mathrm{exit}}\right),$$

where

$$\mathfrak{r}_{\ell m} = \sup_{\zeta \in \mathcal{Q}^{\mathrm{res}}_{\ell m}} \mathsf{ev}^{\mathrm{res}}_{\ell m}(R_{\ell m}, \zeta), \qquad \lambda_{\mathrm{quot}}(q) = \sup_{x \sim_\ell x'} d_q(q(x), q(x'))$$

for a measurable target query $q : X_\ell \to (Q_q, d_q)$, and $\alpha_{\mathrm{exit}} = \sup_{\mu \in \mathcal{Q}^{\mathrm{law}}_{\ell m}} \mathsf{P}_\mu(\tau_{V_m} \le h)$. Here τ_{V_m} is the first exit time from V_m under the declared target dynamics and adapted policy. An important deterministic specialization takes $\mathcal{Q}^{\mathrm{res}}_{\ell m} = D^{\mathrm{eval}}_{\ell m}$ and $\mathsf{ev}^{\mathrm{res}}(R, x) = \|R(x)\|_{Z_m}$. A law-level stochastic residual instead requires its own weak-form evaluator; it is not inserted into the deterministic formula. Each coordinate has its own tolerance and propagation law.

Definition 8.2 (Blackwell order)**.** Experiment E_ℓ Blackwell-dominates E_m, written $\mathsf{E}_\ell \succeq_B \mathsf{E}_m$, when a Markov kernel K satisfies $KP_{\ell,\theta} = P_{m,\theta}$ for every θ. This exact simulation implies $\delta(\mathsf{E}_\ell, \mathsf{E}_m) = 0$. Under the ordinary-kernel infimum used here, the equality $\delta = 0$ by itself means that simulation is arbitrarily accurate; it implies exact Blackwell simulation only when an optimizing kernel exists or an applicable randomization theorem supplies one. The converse deficiency may be positive.

This direction follows Blackwell [1953] and Le Cam [1964], while noting that conventions in the literature vary. Only when needed do we use

$$\Delta(\mathsf{E}, \mathsf{F}) = \max\{\delta(\mathsf{E}, \mathsf{F}), \delta(\mathsf{F}, \mathsf{E})\}.$$

The directed deficiency itself is not a distance.

Proposition 8.3 (Decision-risk consequence)**.** *For a bounded loss $0 \le L \le M$, if $\delta(\mathsf{E}, \mathsf{F}) \le \varepsilon$, then, for every $\eta > 0$, every decision rule for F has a randomized rule based on E whose worst-case risk exceeds it by at most $M(\varepsilon + \eta)$ under the present total-variation convention. If the deficiency infimum is attained and equals ε, one may take $\eta = 0$.*

Proof. Use a channel within η of the deficiency infimum and then apply the target decision rule. The expectation of a function in $[0, M]$ changes by at most $M\|P - Q\|_{\mathrm{TV}}$. □

Definition 8.4 (Boundary relevance and closure screens)**.** For a target Y_{t+h}, boundary data $X_{\partial,t}$, observed interior $X^{\mathrm{obs}}_{\Omega,t}$, exterior $X_{\mathrm{ext},t}$, and legal covariates Z_t, all variables are random elements on a declared experiment probability space. The deployed boundary inputs and legal covariates are $\mathcal{F}_{t-}$-measurable; Y_{t+h} is not, and $X_{\mathrm{ext},t}$ may be a held-out audit variable rather than deployed information. Write $X_{\mathrm{partial},t} := X_{\partial,t}$ for the declared partial boundary record and define

$$G_{\mathrm{partial}} = I(Y_{t+h}; X_{\mathrm{partial},t} \mid Z_t), \qquad S^{\mathrm{aug}}_{\mathrm{ext}} = I(Y_{t+h}; X_{\mathrm{ext},t} \mid X_{\mathrm{partial},t}, X^{\mathrm{obs}}_{\Omega,t}, Z_t).$$

These are, respectively, a relevance screen and an exterior-closure screen for an already augmented information set. If a proposed application predeclares thresholds g_0, s_{ext}, passing $G_{\text{partial}} \geq g_0$ and $S_{\text{ext}}^{\text{aug}} \leq s_{\text{ext}}$ is necessary for that application gate, but is not boundary sufficiency. For a separately declared collection $X_{\text{rest},t}$ of all target-relevant information omitted by $(X_{\text{partial},t}, Z_t)$, exact statistical boundary sufficiency requires $S_{\text{rest}} = 0$. A declared approximate-sufficiency claim instead requires the distinct residual-information condition

$$S_{\text{rest}} = I(Y_{t+h}; X_{\text{rest},t} \mid X_{\text{partial},t}, Z_t) \leq s_{\text{rest}}.$$

All logarithms are base 2, so the screens and thresholds are in bits.

None of these mutual-information conditions establishes causality, a Sobolev trace theorem, a response map, a carrier mechanism, or coefficient reconstruction. A large relevance score cannot compensate for a failed closure or sufficiency condition.

9 Dynamics, recurrence, Koopman operators, and learnability

Let $F : X \to X$ be measurable on a probability space $(X, \mathcal{B}, \mu)$. When F preserves μ, the Koopman operator on $L^2(\mu)$ is the isometry

$$[\mathcal{K}_F g](x) = g(F(x)). \tag{4}$$

It acts on observables, not on states [Koopman, 1931].

Definition 9.1 (Certified approximate mode)**.** For a declared observable space $\mathcal{G} \subseteq L^2(\mu)$, $\phi \in \mathcal{G}$ normalized by $\|\phi\|_{L^2(\mu)} = 1$, and $\lambda \in \mathbb{C}$,

$$\varepsilon_{\text{dyn}}(\phi, \lambda) = \|\mathcal{K}_F \phi - \lambda \phi\|_{L^2(\mu)}.$$

A nonnormalized nonzero mode is assessed by the corresponding relative residual, dividing by $\|\phi\|_{L^2(\mu)}$; $\phi = 0$ is inadmissible. A certificate also states the dictionary, data-access model, norm, sampling law, finite time horizon, limiting order, perturbation class, stability bound, and learnability or Solvability Complexity Index class.

Small training error is not this residual unless the sampling theorem and dictionary-to-operator bridge are proved. A finite compression can generate spectral pollution. Residual DMD supplies a posteriori rejection and convergent spectral procedures for specified information models [Colbrook and Townsend, 2024, Colbrook et al., 2023]. It does not certify arbitrary learned modes.

Definition 9.2 (Recurrence and persistence)**.** Let (Q, d_Q) be Polish, let $p \geq 1$, and let $q_i : \bar{X}_i \to Q$ be declared identity observables. Assume all pushforward laws below lie in $\mathcal{P}_p(Q)$. For each i, a completed admissible fold $F_i : \mathsf{Z}_i^{\text{sc}} = \bar{X}_i \times U_i \times C_i \rightsquigarrow \bar{X}_{i+1}$ and a certified filtration-adapted state-Markov policy $\bar{\pi}_i$ induce the state-marginal kernel

$$P_i(x, A) = \int_{U_i \times C_i} F_i((x, u, c), A)\, \bar{\pi}_i(d(u, c) \mid x).$$

History-dependent policies require history augmentation of $\bar{X}_i$; without that augmentation the displayed Markov kernel is not asserted. Let $\mathbf{X}_0 \sim \mu \in \mathcal{P}(\bar{X}_0)$ and $\mathbf{X}_{i+1} \mid \mathbf{X}_i \sim P_i(\mathbf{X}_i, \cdot)$. Denote the induced path law by P_μ, and define

$$\tau_V = \inf\{i \geq 0 : \mathbf{X}_i \notin V_i\}, \qquad \inf \varnothing = \infty.$$

Endpoint recurrence of a law μ after n folds is

$$\mathrm{Rec}_{q,n}(\mu) = W_{p,Q}((q_0)_\#\mu,\ (q_n)_\#(\mu P_0 \cdots P_{n-1}))\,.$$

Persistent identity through n folds requires, for every prefix $k \le n$,

$$W_{p,Q}((q_0)_\#\mu,\ (q_k)_\#(\mu P_0 \cdots P_{k-1})) \le \varepsilon_k, \qquad \mathsf{P}_\mu(\tau_V \le k) \le \alpha_k^{\mathrm{cum}},$$

where α_k^{cum} is a cumulative exit budget, and every fold's morphism certificate remains admissible. Persistence is therefore stronger than return at the final horizon.

This formalizes, as an organizing principle, the statement: *a persistent object is a recursively maintained finite boundary-state; persistent identity is an operational equivalence maintained under repeated admissible folds across declared scales*. It remains a target-relative definition and conjectural physical principle, not a universal law.

Theorem 9.3 (Prefix error budget)**.** *Suppose the identity discrepancy satisfies $E_{i+1} \le L_i E_i + e_i$, where $L_i \ge 0$ and $e_i \ge 0$ have common units. Then*

$$E_n \le \left(\prod_{j=0}^{n-1} L_j\right) E_0 + \sum_{i=0}^{n-1} e_i \prod_{j=i+1}^{n-1} L_j.$$

At a prefix k, an upper bound not exceeding the declared tolerance certifies the identity coordinate. An upper bound above tolerance is inconclusive: it fails to certify the coordinate but does not prove a violation. Persistence is invalid through the first prefix only when the actual discrepancy, a certified lower bound, or a qualified measurement exceeds tolerance. A later contraction does not erase such a demonstrated excursion.

Proof. Induction on n proves the displayed upper bound. The certification and violation statements follow from interval order and definition 9.2; an upper enclosure that straddles a threshold decides neither side. □

Directed deficiencies satisfy the separate additive bound of theorem 7.7; viability failures satisfy

$$\mathsf{P}(\tau_V \le n) \le \sum_{i=0}^{n} \beta_i$$

without independence whenever the per-step bounds $\mathsf{P}(\mathbf{X}_i \notin V_i) \le \beta_i$ hold. Alternatively the cumulative budget in definition 9.2 is checked directly as $\mathsf{P}(\tau_V \le n) \le \alpha_n^{\mathrm{cum}}$. These coordinates must not be folded into E_n.

9.1 Learnability boundary

The Solvability Complexity Index records how many nested limiting processes a specified computational problem requires and whether error control is available [Colbrook and Hansen, 2023]. Recent Koopman results give matching success and impossibility regimes: for broad declared classes, some spectral tasks require ordered limits, and no universal single-limit learner can solve the stated problems; stronger structure restores certified algorithms [Colbrook et al., 2026]. These are task-class impossibility results, not impossibility of all forecasting.

Convention 9.4 (Koopman claim gate)**.** A recurrence claim based on a learned operator is blocked unless it identifies: (i) discrete eigenvalue, continuous spectrum, pseudospectrum, or spectral measure as the target; (ii) the observable dictionary and its density claim; (iii) the full or sampled residual; (iv) perturbation stability; (v) horizon and sampling law; (vi) the order of limits; and (vii) the applicable positive or negative learnability theorem.

10 Scale transfer, coarse-graining, and ordered limits

Definition 10.1 (Scale-limit certificate)**.** A micro-to-macro claim must state the record

$$(\varepsilon,\ \text{limit order},\ \text{scaling},\ \text{topology},\ \text{boundary conditions},\\ \text{time interval},\ \text{uniformity domain},\ \text{effective parameters},\ \text{error},\ \text{excluded regimes}).$$

If more than one parameter tends to a limit, the certificate records whether the limit is joint, iterated, or path-dependent.

10.1 Certified families and normalized scale profiles

Definition 10.2 (Certified scale family)**.** A certified scale family is a directed comparison family of existing finite BSC systems, recorded as

$$\mathfrak{A} = \left(I, P, \{\lambda_i, \mathsf{S}_i, \mathsf{Cert}_i, A_i, Z_i, L_i, \mathsf{O}_i\}_{i\in I}, \{\mathfrak{M}_{ij}\}_{i\preceq j}\right).$$

Here I is a directed scale index, P is one common topological parameter space, and $\lambda_i \to \infty$ is a dimensionless logarithmic size gauge. Each $(\mathsf{S}_i, \mathsf{Cert}_i)$ is a certified finite system; $\mathfrak{M}_{ij}$ is an interscale comparison or records its defect; $A_i : P \to \mathbb{C}$ is an ideal carrier; $Z_i : P \to \mathbb{C} \setminus \{0\}$ is a normalizer; $L_i = A_i/Z_i$ is the normalized ideal observable; and $\mathsf{O}_i : P \rightsquigarrow Y_i$ is its observation or estimator law.

No categorical identity or composition law is inferred from this record; a strict directed diagram requires those coherences as additional hypotheses. The ideal scalar and the estimator distribution are different typed objects. This comparison family is not a new morphism field, and it does not by itself construct an infinite-system state, quasilocal algebra, limiting dynamics, or physical thermodynamic phase. Such an object requires a separate construction and compatibility theorem.

Definition 10.3 (Normalized scale profile)**.** For a sequence version of definition 10.2, set $X = P$ and write

$$A_N : X \to \mathbb{C}, \qquad Z_N : X \to \mathbb{C} \setminus \{0\}, \qquad L_N = \frac{A_N}{Z_N}, \qquad \lambda_N \longrightarrow \infty.$$

Where the expressions are finite, define

$$\gamma_N(x) = \frac{\log|Z_N(x)|}{\lambda_N}, \qquad \kappa_N(x) = -\frac{\log|A_N(x)|}{\lambda_N}, \qquad \mathcal{R}_N(x) = -\frac{\log|L_N(x)|}{\lambda_N}.$$

The profile declares the limiting carrier A, parameter topology, uniformity domain, error norm, zero convention, and order of limits. If exact finite zeros are admitted, the extended convention $-\log 0 = +\infty$ must be stated. Every logarithm acts on a dimensionless quantity or on a quantity divided by a declared nonzero reference value.

Proposition 10.4 (Normalization collapse)**.** *Fix $x \in X$. If $A_N(x) \to A(x) \in \mathbb{C}$ and $|Z_N(x)| \to \infty$, then $L_N(x) \to 0$. Uniformly on $K \subseteq X$, if $0 < \inf_{x\in K} |Z_N(x)|$ for all sufficiently large N and*

$$\frac{\sup_{x\in K} |A_N(x)|}{\inf_{x\in K} |Z_N(x)|} \longrightarrow 0,$$

then $\sup_K |L_N| \to 0$. At every finite N,

$$\{x : L_N(x) = 0\} = \{x : A_N(x) = 0\}.$$

Proof. Use $|L_N| = |A_N|/|Z_N|$, first pointwise and then with the displayed supremum and infimum. Nonvanishing of Z_N proves the zero-set equality. □

Thus the zero set of the normalized limit may be all of X, while every finite carrier is nonzero. For example, $A_N = 1, Z_N = N$ gives $L_N \to 0$. Conversely $A_N = N, Z_N = N$ gives $L_N = 1$; divergence of the normalizer alone is insufficient.

Theorem 10.5 (Additive scaling signature)**.** *Fix x, assume $A_N(x)$ and $Z_N(x)$ are eventually nonzero, and suppose the finite limits*

$$\gamma(x) = \lim_N \gamma_N(x), \qquad \kappa(x) = \lim_N \kappa_N(x)$$

exist. Then

$$\lim_N \mathcal{R}_N(x) = \gamma(x) + \kappa(x).$$

If $A_N(x) \to A(x) \neq 0$, then $\kappa(x) = 0$. If instead

$$A_N(x) = c(x)e^{-\rho(x)\lambda_N + o(\lambda_N)}, \qquad c(x) \neq 0, \quad \rho(x) \in \mathbb{R},$$

then $\kappa(x) = \rho(x)$.

Proof. The exact identity

$$-\log|L_N| = \log|Z_N| - \log|A_N|$$

gives $\mathcal{R}_N = \gamma_N + \kappa_N$. The remaining conclusions follow by dividing the stated asymptotics by λ_N. □

Existence of the carrier limit does not guarantee an extra rate at its zeros. For $A_N(x) = x + 1/\log N$ and $\lambda_N = \log N$, the limiting carrier vanishes at 0 but $\kappa(0) = 0$. The exponent itself can fail to exist: take $A_N = N^{-1}$ for even N and $A_N = N^{-2}$ for odd N.

Proposition 10.6 (Normalization covariance)**.** *Let $\widetilde{L}_N = c_N L_N$, where $c_N(x) \neq 0$, and suppose both finite rate limits and*

$$\chi(x) = \lim_N \frac{\log|c_N(x)|}{\lambda_N}$$

exist as finite values. Then

$$\widetilde{\mathcal{R}}(x) = \mathcal{R}(x) - \chi(x).$$

If χ is continuous, the two limiting rates have the same discontinuity set. If $\chi = 0$, their values agree.

Proof. Apply

$$-\lambda_N^{-1}\log|\widetilde{L}_N| = -\lambda_N^{-1}\log|L_N| - \lambda_N^{-1}\log|c_N|$$

and take limits. Adding a continuous finite function preserves the discontinuity set. □

A nonzero prefactor always preserves finite zero sets. It preserves limiting rate values only when its scaled logarithm vanishes; a continuous nonzero scaled limit preserves singular locations but shifts rate values.

Proposition 10.7 (Rate stability away from zero)**.** *Let $L_N, \widehat{L}_N \neq 0$ be dimensionless amplitudes at a fixed parameter point. If*

$$|L_N| \geq m_N > 0, \qquad |\widehat{L}_N - L_N| \leq \varepsilon_N < m_N, \qquad \delta_N = \varepsilon_N/m_N,$$

then

$$\left| -\frac{\log|\widehat{L}_N|}{\lambda_N} + \frac{\log|L_N|}{\lambda_N} \right| \leq \frac{-\log(1-\delta_N)}{\lambda_N}.$$

Proof. The triangle inequalities give

$$1 - \delta_N \le \frac{|\widehat{L}_N|}{|L_N|} \le 1 + \delta_N.$$

Take logarithms and use $\log(1+\delta) \le -\log(1-\delta)$ for $0 \le \delta < 1$. □

There is no uniform Lipschitz rate bound as $m_N \downarrow 0$. A small amplitude error can become a large logarithmic-rate error near a zero. Rate transport there requires a lower margin, contour certificate, or root localization rather than an unqualified amplitude tolerance.

Theorem 10.8 (Exceptional-set rate singularities)**.** *Let X be metric, let $\Sigma \subseteq X$ be closed, and let $\gamma, \rho : X \to \mathbb{R}$ be continuous with $\rho > 0$ on $\partial\Sigma$. If*

$$\mathcal{R}(x) = \gamma(x) + \rho(x)\mathbf{1}_\Sigma(x),$$

then

$$\mathrm{Disc}(\mathcal{R}) = \partial\Sigma.$$

If Σ has empty interior, then $\mathrm{Disc}(\mathcal{R}) = \Sigma$*. In particular, when X is a domain in $\mathbb{C}$, this applies if Σ is the zero set of a holomorphic function that is not identically zero on any connected component.*

Proof. Off $\partial\Sigma$, the indicator is locally constant. At a boundary point, approach through Σ and its complement; continuity of γ, ρ and strict positivity of ρ give two limits separated by $\rho(x)$. A closed set with empty interior equals its boundary. The final statement uses the one-variable isolated-zero theorem. □

The empty-interior hypothesis is necessary: for $\Sigma = [-1, 1] \subset \mathbb{R}$, the indicator is discontinuous only at the two endpoints. A discontinuous background can also create unrelated singularities.

Proposition 10.9 (Parameter-slice visibility)**.** *Let $\iota : Q \to X$ be continuous. For every $\mathcal{R} : X \to \mathbb{R}$,*

$$\mathrm{Disc}(\mathcal{R} \circ \iota) \subseteq \iota^{-1}(\mathrm{Disc}(\mathcal{R}))\,.$$

Equality requires a visibility hypothesis ensuring that the slice approaches each branch responsible for the ambient discontinuity.

Proof. Continuity of $\mathcal{R}$ at $\iota(q)$, composed with continuity of ι, implies continuity of $\mathcal{R} \circ \iota$ at q. □

Definition 10.10 (DQPT interpretation certificate)**.** A promotion from a mathematical scale-rate singularity to a dynamical quantum phase transition declares: the finite Hamiltonian or dynamical family; preparation, quench, and control protocols; physical size or volume normalization; amplitude or squared-echo convention; real-time parameter slice; convergence topology and limit order; singularity class (discontinuity, nondifferentiability, failure of real analyticity, or complex nonanalyticity); perturbation regime; estimator and instrument laws; and implementation evidence.

A mathematical discontinuity discharges only the corresponding mathematical item. A singular set in a full complex parameter space need not remain visible on a declared real-time slice, and a family of finite systems is not an undeclared thermodynamic-limit system.

Theorem 10.11 (Certified holomorphic zero-count transfer). *Let Ω be a bounded Jordan domain with $\overline{\Omega} \subset D \subseteq \mathbb{C}$, and let A, A_N be holomorphic on a neighborhood of $\overline{\Omega}$. If certified bounds satisfy*

$$\sup_{\partial\Omega} |A_N - A| \le \varepsilon_N < m_\Omega \le \inf_{\partial\Omega} |A|,$$

then A_N and A have the same number of zeros in Ω, counted with multiplicity.

Proof. The strict boundary inequality is Rouché's theorem. □

Pointwise convergence is insufficient. Even uniform convergence does not stabilize a count when a limiting zero lies on the contour: for the unit disk,

$$A(z) = z - 1, \qquad A_N(z) = z - \left(1 + \frac{(-1)^N}{N}\right),$$

the zero alternates inside and outside.

Theorem 10.12 (Local multiplicity transfer). *Let A be holomorphic near z_0, with a zero of multiplicity m,*

$$A(z) = a_m(z - z_0)^m + O(|z - z_0|^{m+1}), \qquad a_m \neq 0.$$

Let A_N be holomorphic on the same neighborhood and $r_N \downarrow 0$. If

$$\frac{\sup_{|z-z_0|=r_N} |A_N(z) - A(z)|}{|a_m| r_N^m} \longrightarrow q < 1,$$

then A_N has exactly m zeros in $|z - z_0| < r_N$, counted with multiplicity, for all sufficiently large N.

Proof. Uniformly on the shrinking circle, $|A(z)| = |a_m| r_N^m (1 + o(1))$. The hypothesis gives $|A_N - A| < |A|$ there for large N, and Rouché's theorem applies. □

The analytic theorems act on the holomorphic carrier A_N, not on a normalization that depends nonholomorphically on the complex parameter. They certify counts in declared regions, not canonical root pairings or zeros outside those regions. Analytic truncation, implementation, and measurement errors must be enclosed in the same boundary norm before comparison with the margin.

Theorem 10.13 (Periodic elliptic homogenization). *Let $\Omega \subset \mathbb{R}^d$ be bounded and Lipschitz, $Y = (0,1)^d$, and let $A : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ be measurable, symmetric, and Y-periodic, with*

$$\lambda|\xi|^2 \le \xi^\top A(y)\xi \le \Lambda|\xi|^2 \quad \text{a.e. } y, \quad \xi \in \mathbb{R}^d,$$

for fixed $0 < \lambda \le \Lambda < \infty$. For fixed $f \in H^{-1}(\Omega)$, let $u_\varepsilon \in H_0^1(\Omega)$ be the unique weak solution of

$$-\nabla\cdot\big(A(x/\varepsilon)\nabla u_\varepsilon\big) = f.$$

For $j = 1, \dots, d$, let $\chi_j \in H^1_{\mathrm{per}}(Y)/\mathbb{R}$ solve

$$\int_Y A(y)(e_j + \nabla\chi_j)\cdot\nabla v\, dy = 0 \quad \text{for all } v \in H^1_{\mathrm{per}}(Y)/\mathbb{R},$$

and define

$$A_{\mathrm{hom}} e_j = |Y|^{-1} \int_Y A(y)(e_j + \nabla\chi_j)\, dy.$$

As $\varepsilon \downarrow 0$, after A, Ω, f are fixed,

$$u_\varepsilon \rightharpoonup u_0 \text{ in } H_0^1(\Omega), \qquad u_\varepsilon \to u_0 \text{ in } L^2(\Omega),$$

and

$$A(x/\varepsilon)\nabla u_\varepsilon \rightharpoonup A_{\mathrm{hom}}\nabla u_0 \quad \text{in } L^2(\Omega;\mathbb{R}^d),$$

where $u_0 \in H_0^1(\Omega)$ solves $-\nabla\cdot(A_{\mathrm{hom}}\nabla u_0) = f$.

Proof. Uniform ellipticity and Lax–Milgram give a uniform H_0^1 bound. Weak compactness and Rellich compactness supply subsequential weak H_0^1 and strong L^2 limits. Periodic corrector test functions in the weak formulation identify every subsequential flux limit with $A_{\mathrm{hom}}\nabla u_0$; uniqueness of the effective weak solution removes subsequence dependence. This is the standard corrector argument under the displayed hypotheses [Bensoussan et al., 1978]. □

This stationary theorem has no time limit. It supplies no quantitative rate under the stated hypotheses and no uniformity over changing coefficient classes, data, or domains. Degenerate, nonlinear, random, oscillating-boundary, and coupled time-dependent regimes are excluded. Stochastic homogenization similarly exposes stationarity, mixing, dimension, and ellipticity assumptions [Gloria and Otto, 2012, Armstrong and Smart, 2016]. This is the model for BSC scale claims: no universal rate is inherited from the word "coarse-graining."

Definition 10.14 (Dynamical scale defect)**.** Let $(\bar{X}_b, d_b)$ be Polish. For a measurable state map $R^X_{ab} : \bar{X}_a \to \bar{X}_b$, source and target Markov kernels P_a, P_b, and an input law μ such that both displayed laws lie in $\mathcal{P}_p(\bar{X}_b)$, define

$$T^{\mathrm{dyn}}_{ab}(\mu) = W_p\Big((R^X_{ab})_\#(\mu P_a), ((R^X_{ab})_\#\mu)P_b\Big).$$

Proposition 10.15 (Iterated scale defect)**.** *Suppose P_a, P_b preserve the required p-moments,*

$$T^{\mathrm{dyn}}_{ab}(\mu P_a^j) \le \tau \quad \textit{for } j = 0, \ldots, n-1,$$

and P_b is L_b-Lipschitz in the Wasserstein distance induced by d_b on every propagated target law. Then

$$W_p\Big((R^X_{ab})_\#(\mu P_a^n), ((R^X_{ab})_\#\mu)P_b^n\Big) \le \tau \sum_{k=0}^{n-1} L_b^k.$$

Proof. Insert the one-step commuting square at each time and iterate the Wasserstein triangle and Lipschitz inequalities. □

Classical kinetic-to-fluid limits exhibit the necessary scope [Bardos et al., 1993, Golse and Saint-Raymond, 2004]. The 2025 hard-sphere program of Deng et al. [2025] is a preprint result for a specified hard-sphere → Boltzmann → Euler/Navier–Stokes–Fourier chain on specified tori and ordered regimes, together with the long-time hard-sphere derivation in Deng et al. [2024]. It is not a derivation of all continuum mechanics or a settlement of every reading of Hilbert's sixth problem. BSC uses it as a case in disciplined scope, not as a universal bridge.

11 Local-to-global consistency and sheaf obstruction

Let $\mathcal{U}$ be a base of contexts closed under the finite intersections and relevant unions used below, or replace it by the topology generated by the declared cover. More generally, the same definitions may be made on a site after its Grothendieck topology is specified.

Definition 11.1 (Observation sheaf)**.** An observation sheaf $\mathscr{F}$ assigns to every context $U \in \mathcal{U}$ a set, vector space, or probability model $\mathscr{F}(U)$, and to $V \subseteq U$ a restriction $\rho^U_V : \mathscr{F}(U) \to \mathscr{F}(V)$, with identity and composition laws. Local sections $s_i \in \mathscr{F}(U_i)$ are compatible when

$$\rho^{U_i}_{U_i \cap U_j} s_i = \rho^{U_j}_{U_i \cap U_j} s_j$$

for every nonempty overlap. The sheaf axiom states that compatible sections have a unique global section $s \in \mathscr{F}(\bigcup_i U_i)$ restricting to them.

The last statement is an axiom of a chosen sheaf, not a fact about every data assignment. In contextual models one instead asks whether a compatible family or local relation admits a global section; failure is the obstruction [Abramsky and Brandenburger, 2011].

Definition 11.2 (Consistency operators)**.** For a finite cell complex with finite-dimensional inner-product stalks, let C^0 be vertex assignments, C^1 edge discrepancy spaces, and $d^0 : C^0 \to C^1$ the signed consistency difference constructed from the cellular sheaf's coface maps. This cellular coface convention is covariant along face incidence and must not be confused with the contravariant restriction maps of the ordinary sheaf in definition 11.1. The sheaf Laplacian is

$$L_{\mathscr{F}} = (d^0)^* d^0.$$

Then $\ker L_{\mathscr{F}} = \ker d^0$ is the space of exactly consistent global assignments. For data $s \in C^0$, $\|d^0 s\|$ is a declared consistency residual. Its scale depends on the chosen inner products.

This construction follows the finite cellular hypotheses of Hansen and Ghrist [2019]; it is not a canonical scalar for arbitrary heterogeneous data. For torsor-valued or parity constraints, a cocycle class in H^1 may obstruct gluing, but cohomology is invoked only after the coefficient system and differential have been constructed. The corpus's earlier "scale cohomology" lacked these objects and is here demoted to path or naturality defect.

Proposition 11.3 (Gluing gate)**.** *For an actual sheaf $\mathscr{F}$, pairwise compatible local sections glue uniquely. For a presheaf, relation-valued assignment, or noisy family, local nonemptiness alone does not imply a global section.*

Proof. The first statement is the sheaf axiom. The second is witnessed by worked fixture 16.6. □

12 Open-system and non-invertible composition

12.1 A concrete interface category

Let $\mathcal{T}$ be a set of port types, including units and direction, and let $\mathsf{I}_{\mathcal{T}}$ be the full subcategory of $\mathrm{Set}/\mathcal{T}$ on maps $p : A \to \mathcal{T}$ with finite domain. Thus it is the category of finite typed interfaces even when $\mathcal{T}$ itself is infinite. Let C be a category of internal carriers with finite colimits, and let $L : \mathsf{I}_{\mathcal{T}} \to \mathsf{C}$ embed an interface as a free boundary carrier.

Assumption 12.1 (Conditional decoration schema)**.** Fix one mode $\mathsf{r} \in \{\mathrm{det}, \mathrm{stoch}, \mathrm{nondet}\}$ and one compatible time object $\mathbb{T}$. Assume that a lax symmetric monoidal functor

$$\mathcal{D}_{\mathsf{r},\mathbb{T}} : (\mathsf{C}, +, 0) \longrightarrow (\mathrm{Set}, \times, 1)$$

with laxator $\varphi_{N,M} : \mathcal{D}(N) \times \mathcal{D}(M) \to \mathcal{D}(N + M)$ and unit $\varphi_0 : 1 \to \mathcal{D}(0)$. An element of $\mathcal{D}(N)$ is a record

$$d_N = (\widehat{S}_N, \mathsf{r}, \mathbb{T}, \mathsf{Beh}_N, \mathsf{Cert}_N),$$

where Beh_N is respectively a map/flow, Markov kernel, or trajectory relation with typed boundary ports. The functorial action $\mathcal{D}(f)$ specifies how behavior and certificates are transported along a carrier map f. Illegal unit, clock, boundary, or resource identifications map to a distinguished absorbing failed decoration and are blocked by admissibility. Mixed modes or clocks require a separately declared lax monoidal comparison functor; they are not composed by default.

Definition 12.2 (Typed open BSC system)**.** A deterministic, stochastic, or nondeterministic open system of the fixed mode and clock from A to B is an isomorphism class of structured cospans

$$L(A) \xrightarrow{i} N \xleftarrow{o} L(B)$$

decorated by $d_N \in \mathcal{D}_{\mathsf{r},\mathbb{T}}(N)$.

Given $A \to N \leftarrow B$ and $B \to M \leftarrow C$, form the pushout $P = N +_{L(B)} M$ in C, with coprojections $j_N : N \to P$, $j_M : M \to P$. The composite is

$$L(A) \longrightarrow N +_{L(B)} M \longleftarrow L(C),$$

with decoration

$$d_P = \mathcal{D}([j_N, j_M])\big(\varphi_{N,M}(d_N, d_M)\big).$$

The identity on A is $L(A) \xrightarrow{\mathrm{id}} L(A) \xleftarrow{\mathrm{id}} L(A)$, decorated by

$$d_A^{\mathrm{id}} = \mathcal{D}(0 \to L(A))(\varphi_0(*)).$$

Theorem 12.3 (Interface category)**.** *If* C *has chosen finite colimits,* L *is a left adjoint and strong symmetric monoidal for finite coproducts, and* $\mathcal{D}_{\mathsf{r},\mathbb{T}}$ *is the decoration datum of assumption 12.1, then the objects of* $\mathsf{I}_{\mathcal{T}}$ *and morphisms of definition 12.2 form a category, up to the standard isomorphism classes. Disjoint union supplies a symmetric monoidal product when the decorating semantics is monoidal.*

Proof. Pushout associativity and the universal property give associativity up to canonical isomorphism; passing to isomorphism classes makes composition associative. Naturality and the associativity and unit axioms of the laxator give the same laws for d_P. Identity cospans and φ_0 give the unit laws. □

This theorem gives a generic open-system construction conditional on a supplied functor $\mathcal{D}_{\mathsf{r},\mathbb{T}}$. It does not prove that the eight-field BSC morphism records canonically define such a functor, nor that their certificate transport satisfies the laxator coherence laws. Constructing that BSC-specific decoration and proving its coherence remain open. The conditional schema is a hybrid in which structured cospans carry behavior and certificate decorations. Decorated and structured cospans are distinct constructions, with precise comparison hypotheses [Fong, 2015, Baez and Courser, 2020, Baez et al., 2022]. Heterogeneous internal types remain in the apex decoration; only declared ports are identified. Stochastic behavior may also be organized in a Markov category, where copying and discarding are structural but disintegration is not automatically available [Fritz, 2020]. Accordingly, equality-gluing of stochastic ports is legal here only when the chosen $\mathcal{D}_{\mathrm{stoch},\mathbb{T}}$ explicitly supplies that composition; no conditional distribution or disintegration is inferred from the cospan alone.

12.2 Non-invertible transports and defects

Definition 12.4 (Fusion morphism)**.** Let $(\mathscr{D}, \otimes, \mathbf{1}, \alpha, \lambda, \rho)$ be a k-linear additive monoidal category with finite biproducts. A finite fusion record is a finite family of defect objects $D_a \in \mathscr{D}$, including $D_{\mathbf{1}} \simeq \mathbf{1}$, nonnegative integers N_{ab}^{c} of finite support, and specified isomorphisms

$$D_a \otimes D_b \simeq \bigoplus_c D_c^{\oplus N_{ab}^{c}},$$

coherent with the associator and unit constraints; equivalently their two triple-fusion composites agree under the pentagon. If the defects act on a physical system A, the record additionally supplies a monoidal functor $\mathscr{D} \to \mathrm{End}(A)$ in the declared ambient bicategory. Vector-space or lower-dimensional-TFT multiplicities require an explicitly tensored higher category and are a separate variant, not an untyped replacement for the integers above.

A quotient morphism pulls observables back only to the quotient-invariant subalgebra. A stochastic channel is generally irreversible and is compared by deficiency. A correspondence needs selection semantics. A fusion morphism branches, and the absence of an inverse is structural, not a failed group axiom. Physical use requires construction of the QFT, defect category, fusion, anomaly, and boundary conditions; schematic fusion does not validate a duality [Gaiotto et al., 2015, Choi et al., 2022, Apruzzi et al., 2023]. Symmetry-TFT constructions further organize non-invertible defect data in specified field theories; they do not supply a generic duality certificate [Kaidi et al., 2023].

13 Topological quantities and physical charge

Definition 13.1 (Topological charge)**.** Let A be an abelian coefficient group. For $[\omega] \in H^k(M; A)$ and $[C] \in H_k(M; \mathbb{Z})$, the Kronecker evaluation is

$$Q_{\text{top}} = \langle [\omega], [C] \rangle \in A.$$

It is invariant under the stated homology/cohomology equivalences. In the differential-form case, $A = \mathbb{R}$, ω is a closed k-form, and the de Rham pairing is $Q_{\text{top}} = \int_C \omega$, with any normalization stated separately.

Definition 13.2 (Topology-to-physical-charge bridge)**.** When a claim asserts that a physical charge is determined by Q_{top}, it must supply a typed map

$$\chi : \mathrm{Dom}(\chi) \subseteq A \longrightarrow \mathcal{Q}_{\text{phys}}, \qquad q_{\text{phys}} = \chi(Q_{\text{top}}),$$

with $Q_{\text{top}} \in \mathrm{Dom}(\chi)$ and $\mathcal{Q}_{\text{phys}}$ a declared physical quantity space with charge units. The map must be derived from at least one declared physical structure: gauge representation and coupling, an action and Noether current, a Gauss-law boundary flux, anomaly constraint, boundary condition, or calibrated measurement protocol. Its normalization and units are part of the derivation. This is an obligation on topology-to-charge claims, not a definition of every physical charge.

Proposition 13.3 (Normalization underdetermination)**.** *Let the only supplied carrier data be a winding invariant $Q_{\text{top}} \in \mathbb{Z}$. These data do not identify a real-valued physical-charge map: they are compatible with every normalization $\chi_c : \mathbb{Z} \to \mathbb{R}$, $\chi_c(n) = cn$, $c \in \mathbb{R}$.*

Proof. For every real constant c, $\chi_c(n) = cn$ is a group homomorphism. The supplied data select none of the continuum of normalizations, nor the electric unit, sign convention, matter representation, or coupling to an electromagnetic current. A nonlinear or quotient-valued bridge introduces still more choices. □

Winding can be a valid topological carrier while the physical promotion is blocked. Modern monopole–center-vortex work illustrates the full price of a bridge: it specifies $SU(N)$ gauge theory, compactification, 't Hooft flux, fractional topological charge, semiclassical action and fugacity, vacuum branches, anomaly constraints, and representation-sensitive string tensions in a controlled regime [Tanizaki and Ünsal, 2022, Hayashi and Tanizaki, 2024]. The adiabatic passage to strongly coupled undeformed $\mathbb{R}^4$ remains conjectural in that work; it does not provide hadron spectroscopy or validate an unrelated toroidal model.

14 Certificates, claim ledgers, and demotion

Definition 14.1 (Certificate schema)**.** A certificate is a machine-readable and human-inspectable record

$$\mathsf{Cert} = (\,\mathsf{id}, \mathsf{claim}, \mathsf{types}, \mathsf{assumptions}, \mathsf{tolerances}, \mathsf{sources},$$
$$\mathsf{artifacts}, \mathsf{hashes}, \mathsf{proof}, \mathsf{execution}, \mathsf{obligations}, \mathsf{Stat}).$$

The proof field names the proposition and proof object or says "mathematical proof in text." The execution field records code, inputs, environment, command, output, tolerances, and receipt, or says "unexecuted." Hashes identify artifacts; they do not establish truth.

Definition 14.2 (Mechanical verification)**.** A claim is mechanically verified only when a declared trusted kernel accepts a proof object for the encoded proposition in a pinned environment. The certificate records the kernel and prover version, theorem identifier, definitions, imported axioms, dependency hashes, build command, and accepted output. Kernel acceptance proves the encoded statement relative to those foundations; it does not prove that the encoding represents the intended physical system.

Lean supplies a small-kernel architecture and explicit proof objects [de Moura and Ullrich, 2021]; large formalizations show the value of modular dependency control [Hales et al., 2017, Commelin and Topaz, 2024]. A blueprint or proof DAG is not itself a proof.

Definition 14.3 (Numerical and empirical status)**.** A numerical claim requires actual execution with code, inputs, environment, output, convergence and tolerance criteria, and hashes. A model description without execution is computationally unexecuted. An empirical claim also requires a sampling frame, instrument and calibration, protocol, exclusions, uncertainty analysis, and data provenance. Simulation agreement is not structural equivalence.

14.1 Simulation evidence profile and deployment compatibility

Several neighboring uses of "simulation" must not be conflated. In definitions 7.1 and 8.2, *statistical simulation* means a Markov kernel comparing experiments in the Blackwell–Le Cam sense. *Computational simulation* means numerical execution of a declared mathematical model by a solver. A learned *surrogate* is a fitted replacement for a declared component, while *in-simulator deployment* inserts that component into a host solver, controller, or coupled pipeline. None of these meanings inherits the evidence of another without a typed bridge.

Definition 14.4 (Simulation evidence profile)**.** For a claim c and evidence identity ι, a simulation evidence profile is the certificate subrecord

$$\mathsf{SEC}_{c,\iota} = \left(\mathcal{U}_c, I_c^{\ell}, J_c^{\ell}, J_c^{g}, \{\mathsf{E}_{c,i}\}_{i\in I_c^{\ell}}, \{g_{c,k}\}_{k\in J_c^{g}}, \Phi_c, \boldsymbol{\rho}_c, \mathsf{Prov}_{c,\iota}\right).$$

It augments $\mathsf{Cert}_{\ell m}$; it is not a ninth field of $\mathfrak{M}_{\ell\to m}$. The intended-use record

$$\mathcal{U}_c = (D_c, H_c, Q_c, \pi_c, \mathsf{BC}_c, \mathsf{Units}_c, \tau_c)$$

fixes the operating domain, horizon, quantity of interest, intervention or control policy, initial and boundary conditions, units, and claim tolerances. I_c^{ℓ} indexes typed source evidence coordinates, J_c^{ℓ} indexes applicable coordinates of the existing BSC loss vector, and J_c^{g} indexes Boolean hard gates with values true, false, or unevaluated. With ordered source and target products $\mathcal{V}_c = \prod_{i\in I_c^{\ell}} V_{c,i}$ and $\mathcal{W}_c = \prod_{j\in J_c^{\ell}} W_{c,j}$, the declared monotone, unit-respecting map $\Phi_c : \mathcal{V}_c \to \mathcal{W}_c$ has a proved relation $\boldsymbol{\ell}_c^0 \le_{\mathcal{W}_c} \Phi_c(\boldsymbol{\eta}_c)$. It is the typed bridge from source estimands to BSC loss coordinates.

For every statistically evaluated source coordinate i, the evidence record is

$$\mathsf{E}_{c,i} = \left(\eta_{c,i}, d_{c,i}, \mathcal{O}_{c,i}, \widehat{\eta}_{c,i}, n_{c,i}, \alpha_{c,i}, [L^{\mathrm{src}}_{c,i}, U^{\mathrm{src}}_{c,i}], \varepsilon^{\mathrm{opt}}_{c,i}, \Psi_{c,i}\right).$$

It names the estimand, metric or divergence, sampling or oracle model, estimator, sample count, coverage-failure probability, confidence or deterministic enclosure, optimization-gap certificate, and a proved proxy relationship $\Psi_{c,i}$, or explicitly records that none is known. All records used together live on one declared joint observation-and-analysis space $(\Omega_c, \mathcal{F}_c, \mathsf{P}_c)$, with measurable random endpoints. The profile must justify

$$\mathsf{P}_c\Big\{\eta_{c,i} \in [L^{\mathrm{src}}_{c,i}, U^{\mathrm{src}}_{c,i}] \text{ for every } i \in I^{\ell}_c\Big\} \geq 1 - \alpha_c.$$

Separate marginal coverage statements do not imply this display; a simultaneous procedure or the marginal guarantees $\mathsf{P}_c\{\eta_{c,i} \notin [L^{\mathrm{src}}_{c,i}, U^{\mathrm{src}}_{c,i}]\} \leq \alpha_{c,i}$ together with $\sum_i \alpha_{c,i} \leq \alpha_c$ is required. Data reuse, adaptive search, calibration, and dependence belong to the joint law. Writing

$$\boldsymbol{U}^{\mathrm{src}}_c = (U^{\mathrm{src}}_{c,i})_i, \qquad \boldsymbol{U}^0_c = \Phi_c(\boldsymbol{U}^{\mathrm{src}}_c) = (U^0_{c,j})_j,$$

monotonicity gives $\boldsymbol{\ell}^0_c \leq \Phi_c(\boldsymbol{\eta}_c) \leq \boldsymbol{U}^0_c$ on this event. Setting $\alpha_{c,i} = 0$ asserts probability-one coverage; determinism additionally requires pointwise deterministic endpoints and inequalities.

The evidence identity is factored rather than represented by one opaque label:

$$\iota = (\iota_{\mathrm{cand}}, \iota_{\mathrm{data}}, \iota_{\mathrm{analysis}}, \iota_{\mathrm{env}}, \iota_{\mathrm{contract}}).$$

These factors bind, respectively, the candidate model, simulator, solver and surrogate; training, calibration, validation and reference data; estimator, optimization, analysis code and seeds; runtime, hardware, services and clock; and intended use, thresholds, metrics, gates and evaluation policy. Each is the root of its claim-relevant dependency graph. A relevant change creates a new evidence identity; a prose-only repository change need not. Exact transfer requires equality of every factor on which the evidence depends. Otherwise, evidence transfers only through a theorem quantified over both identities or a freshly certified compatibility record $\Gamma_{\iota \to \iota'}$.

For deficiency, the comparison and implementation problems remain distinct. For a tested channel $\widehat{K}$,

$$\delta(\mathsf{E}_\ell, \mathsf{E}_m) \leq e_{\ell m}(\widehat{K}).$$

Thus a valid upper enclosure for $e_{\ell m}(\widehat{K})$ is also an upper enclosure for the ordinary deficiency, but it is not an estimate of the infimum and gives no lower bound. Reporting an optimization gap requires a certificate such as

$$L_\delta \leq \delta(\mathsf{E}_\ell, \mathsf{E}_m) \leq e_{\ell m}(\widehat{K}) \leq U_e, \qquad U_e - L_\delta \leq \omega^{\mathrm{opt}}.$$

An RMSE, MMD, Wasserstein, or other proxy may replace neither total variation nor deficiency unless $\Psi_{c,i}$ proves the needed direction and constants on the declared distribution class. The sampling or oracle model is substantive: ordinary-sample estimation of high-dimensional total variation can have exponential sample complexity, while stronger conditional oracles define a different information model [Meel et al., 2025]. For example, $P = \delta_0$ and $Q = \delta_\epsilon$ on $\mathbb{R}$, with $\epsilon > 0$, satisfy

$$W_1(P, Q) = \epsilon \qquad \text{but} \qquad d_{\mathrm{TV}}(P, Q) = 1.$$

Thus an arbitrarily small Wasserstein value does not alone certify a small total-variation coordinate. This also fails for deficiency: take $\Theta = \{-1, +1\}$, $P_{-1} = P_{+1} = \delta_0$, and $Q_{\pm 1} = \delta_{\pm\epsilon}$. Then $\sup_\theta W_1(P_\theta, Q_\theta) = \epsilon$, but every source channel produces one common law R. The total-variation triangle inequality gives $\max_\theta \|R - Q_\theta\|_{\mathrm{TV}} \geq 1/2$, and the equal mixture of the two target atoms attains equality. Hence $\delta(\mathsf{E}, \mathsf{F}) = 1/2$.

Proposition 14.5 (Compatibility-bounded deployment)**.** *Fix a claim c, a frozen evidence identity f, and a deployment identity d. Suppose that, jointly for every required loss coordinate $j \in J_c^\ell$,*

$$\ell^0_{c,j} \le U^0_{c,j}, \qquad \ell^1_{c,j} \le \ell^0_{c,j} + \rho_{c,j}, \qquad \rho_{c,j} = \rho^{\mathrm{op}}_{c,j} + \rho^{\mathrm{ver}}_{c,j} + \rho^{\mathrm{cpl}}_{c,j},$$

where all terms have the units of coordinate j. They bound, respectively, operating-region or distribution drift, version or pipeline drift, and coupling or clock exposure introduced by the declared change. Frozen-loss estimator uncertainty belongs inside $U^0_{c,j}$ and is not counted again in $\rho_{c,j}$. Uncertainty in estimating deployment change is counted exactly once, either in the compatibility reserve on the joint event or as a separate source coordinate propagated into that reserve. If

$$U^0_{c,j} + \rho_{c,j} \le \tau_{c,j} \quad \textit{for every } j \in J_c^\ell,$$

then the deployment satisfies the loss-coordinate part of definition 7.2. If it is also well formed, every hard gate is true, and readiness meets Req_c, it is admissible for c. For a stochastic compatibility certificate the conclusion holds on its declared joint-coverage event, whose probability is at least its declared lower bound.

Proof. On the certified event, for every $j \in J_c^\ell$,

$$\ell^1_{c,j} \le \ell^0_{c,j} + \rho_{c,j} \le U^0_{c,j} + \rho_{c,j} \le \tau_{c,j}.$$

Every coordinate of the BSC loss vector is nonnegative, so on this event

$$I^1_{c,j} = [0, U^0_{c,j} + \rho_{c,j}]$$

is a proved deployment enclosure containing $\ell^1_{c,j}$. Thus the deployment is evaluated for c. Conjunction with well-formedness, the hard gates, and the readiness requirements is precisely claim-relative admissibility. The joint-coverage statement controls the probability that all displayed inequalities hold simultaneously. A probability-one conclusion obtained by setting $\alpha_c = 0$ is deterministic only when the enclosures and compatibility inequalities hold pointwise (in particular, on a singleton probability space). $\square$

The certified remaining reserve is

$$s_{c,j} = \tau_{c,j} - (U^0_{c,j} + \rho_{c,j}).$$

Equality in the headroom test can establish admissibility under the complete frozen bounds, but $s_{c,j} = 0$ leaves no certified margin for any additional positive perturbation. A new simulator, solver, dataset, analysis cycle, surrogate, or pipeline hash starts a new evaluation unless $\Gamma_{f\to d}$ bounds its effect on every required coordinate. A new hash establishes a new identity, not compatibility; an unchanged hash does not bound operating-region drift.

Corollary 14.6 (Coupled-surrogate prefix specialization)**.** *For host h, suppose a standalone surrogate has a certified one-step interface error at most ε_h, and the coupled host error satisfies*

$$E^{(h)}_{k+1} \le g_h E^{(h)}_k + \varepsilon_h, \qquad E^{(h)}_0 = 0, \qquad g_h \ge 0.$$

Then at every prefix n,

$$E^{(h)}_n \le \varepsilon_h \sum_{r=0}^{n-1} g_h^r = \begin{cases} \varepsilon_h(1-g_h^n)/(1-g_h), & g_h \ne 1, \\ n\varepsilon_h, & g_h = 1. \end{cases}$$

Consequently, two hosts can have the same standalone surrogate error and different loss-coordinate tolerance outcomes solely because their certified coupling gains or horizons differ.

Proof. Apply the prefix error budget of theorem 9.3 with $L_k = g_h$ and $e_k = \varepsilon_h$, then sum the geometric series. The coordinate statement follows by comparing each required prefix enclosure with its claim tolerance. Full claim-relative admissibility still requires all other applicable coordinates, gates, and readiness obligations. □

Remark 14.7 (Exact stable-host Fixture F10). Fixture F10 makes the scalar recurrences in corollary 14.6 exact with the same standalone interface error $\varepsilon = 1/100$, horizon 10, and host-state tolerance $\tau = 1/20$. Host A has $g_A = 1/2$, so its largest prefix error is

$$E_{10}^{(A)} = \frac{1}{100}\sum_{r=0}^{9}\left(\frac{1}{2}\right)^r = \frac{1023}{51200} < \frac{1}{20}.$$

Host B has $g_B = 9/10$, so

$$E_{10}^{(B)} = \frac{1}{100}\sum_{r=0}^{9}\left(\frac{9}{10}\right)^r = \frac{6513215599}{100000000000} > \frac{1}{20}.$$

Both gains are strictly below one. Host A remains within this tolerance at every prefix, whereas Host B first violates it at step 7, where its exact actual error is $5217031/100000000 > 1/20$. The retained deterministic CPython receipt uses exact rational arithmetic, binds claim identifier BSC-FIX-10, maps the candidate, data, analysis, environment, and contract identity factors (with data typed not applicable), and records the initial condition and both exact state trajectories. The checker independently validates the arithmetic and byte-for-byte regeneration; the repository manifest externally binds the receipt hash, avoiding self-reference. This is an exact execution receipt for the declared finite recurrence and one loss coordinate only, not full BSC admissibility, physical validation, a general surrogate guarantee, or evidence for a Riemann claim.

Prior-art boundary. The separation of verification, validation, uncertainty, intended use, and decision authority is established modeling-and-simulation practice, not a BSC invention. NASA's active standard and companion handbook organize credibility across the model life cycle and use context [National Aeronautics and Space Administration, 2026]; contemporary SciML guidance separately treats problem definition, code and solution verification, calibration, validation, application domains, and continuous credibility [Jakeman et al., 2025]. Finite-horizon in-simulator surrogate bounds already expose coupling sensitivity and horizon dependence [Ellinas et al., 2026], while ECMWF's reported response to an upstream IFS-cycle change provides an operational example of version-conditioned performance, reverification, and operational response [Ben Bouallègue et al., 2026]. The BSC-specific contribution here is narrower: bind those established obligations to one claim's typed loss coordinates, joint coverage, factored evidence identity, compatibility record, readiness caps, and prefix-admissibility rule. No solver improvement, surrogate accuracy gain, speedup, or empirical validation is claimed by this profile.

14.2 Operational report-channel core

A second certificate refinement applies when heterogeneous physical systems are compared only at a common report boundary. The shared structure is

$$\theta \longmapsto \mathsf{Prep}_\theta \longmapsto \mathsf{Drive}_\theta \longmapsto \mathsf{Measure}_\theta \longmapsto \mathsf{Report}_\theta \longmapsto c_\theta.$$

It does not identify microscopic dynamics. It makes preparation, drive, measurement, reporting and decision into separately falsifiable interfaces.

Definition 14.8 (Operational report envelope)**.** Let Θ be a parameter or intervention space. An operational report envelope is

$$\mathsf{ORE} = \Big(\Theta, \{z_0(\theta)\}_{\theta\in\Theta}, \{T_k\}_{k=1}^m, Z, \{P_\theta^Z\}_{\theta\in\Theta}, \mathsf{Cert}_{\mathrm{ORE}}\Big).$$

Each z_k is a classical law or finite-dimensional density operator. Classical stages are Markov kernels, quantum stages are CPTP maps, quantum-to-classical stages are declared POVM channels, and the terminal P_θ^Z is a classical report law. Every control, clock, memory, domain, measurement, report rule and evidence identity on which the claim depends belongs in the certificate or augmented state.

Write $d_k = d_{\mathrm{TV}}$ on a classical interface and

$$d_k = D_{\mathrm{tr}}, \qquad D_{\mathrm{tr}}(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1,$$

on a quantum interface. Let $T_k, \widehat{T}_k$ be ideal and implemented stages with the same typed interfaces, and put

$$z_k = T_k z_{k-1}, \qquad \widehat{z}_k = \widehat{T}_k \widehat{z}_{k-1}, \qquad E_k = d_k(\widehat{z}_k, z_k).$$

Theorem 14.9 (Compatible fixed-interface channel bound)**.** *Suppose*

$$d_0(\widehat{z}_0, z_0) \le \varepsilon_0.$$

For every stage k, let $\widehat{\mathcal{R}}_{k-1}$ contain every implemented reachable input under the declared preparation, controls, horizon and prior defects. Assume

$$d_k(\widehat{T}_k\widehat{z}, T_k\widehat{z}) \le \varepsilon_k \quad \textit{for every } \widehat{z}\in\widehat{\mathcal{R}}_{k-1},$$

and, on every reachable ideal/implemented pair,

$$d_k(T_k\widehat{z}, T_k z) \le \eta_k d_{k-1}(\widehat{z}, z).$$

Then

$$E_k \le \varepsilon_k + \eta_k E_{k-1}$$

and

$$\boxed{E_m \le \sum_{k=0}^m \varepsilon_k \prod_{j=k+1}^m \eta_j}. \tag{5}$$

Proof. At the implemented input $\widehat{z}_{k-1}$, the triangle inequality gives

$$\begin{aligned} E_k &\le d_k(\widehat{T}_k\widehat{z}_{k-1}, T_k\widehat{z}_{k-1}) \\ &\quad + d_k(T_k\widehat{z}_{k-1}, T_k z_{k-1}) \le \varepsilon_k + \eta_k E_{k-1}. \end{aligned}$$

Iterating gives (5), with empty product 1. □

Since both metrics are at most 1, the right-hand side of (5) may be clipped at 1. Estimated ε_k or η_k require a declared confidence event and simultaneous-coverage rule; the recurrence itself creates no coverage.

A local defect certified only on the ideal reachable set is insufficient: the first term in the proof is evaluated at an implemented reachable input. For Markov kernels, a Dobrushin coefficient can supply η_k. CPTP trace-distance contractivity supplies $\eta_k \le 1$; strict quantum contraction requires a separate global or restricted theorem. For a fixed POVM $\mathcal{M}$,

$$d_{\mathrm{TV}}(\mathcal{M}(\rho), \mathcal{M}(\sigma)) \le D_{\mathrm{tr}}(\rho,\sigma),$$

and subsequent report kernels contract total variation. A changed measurement or report is another defective stage. A complete heralding instrument retaining success and failure is CPTP, but normalization on the successful branch is nonlinear and may amplify distinguishability. A conditioned report therefore needs a separate conditioning bound and a certified positive lower bound on success probability.

Theorem 14.9 is ledger claim BSC-CHN-01. It is a fixed-interface state/law corollary of BSC-SIM-03 and data processing. It does not compose partial domains, observable pullbacks, equation residuals, deficiencies, completions or certificate witnesses in the full eight-field record. In particular it does not resolve BSC-QOP-03.

Proposition 14.10 (No downstream resurrection and identified sets)**.** *Classical report kernels and CPTP channels cannot increase total-variation or trace-distance distinguishability. Consequently, if two parameters induce the same terminal report law, no decoder using only that report can distinguish them. For a deterministic forward report $F : \Theta \to Y$,*

$$\mathcal{I}_\varepsilon(y) = \{\theta : d_Y(F(\theta), y) \le \varepsilon\}$$

is the identified set. An exact target property q is licensed by that report only when it is constant on the applicable identified set; at zero error this is precisely constancy on fibers of F. For a declared center rule $c : Y \to Q$, a radius-τ claim additionally requires

$$\sup_{\theta \in \mathcal{I}_\varepsilon(y)} d_Q(q(\theta), c(y)) \le \tau.$$

If the center may instead be chosen after fixing the identified set, the exact condition is the corresponding infimum over $c \in Q$. Diameter at most 2τ is necessary but not sufficient in a general metric space.

Proof. The contraction statements are the classical and quantum data-processing inequalities. Equality of terminal laws gives zero distinguishability. Constancy on F-fibers is equivalent to the existence of a factorization $q = \bar{q} \circ F$, as in proposition 6.4. □

This is ledger claim BSC-CHN-02.

Proposition 14.11 (Forward error encloses compatible inverse sets)**.** *Let deterministic actual and ideal reports $F, G : \Theta \to Y$ obey $\sup_\theta d_Y(F(\theta), G(\theta)) \le B$, and put*

$$\mathcal{I}_F(y, \delta) = \{\theta : d_Y(F(\theta), y) \le \delta\}.$$

Then

$$\boxed{\mathcal{I}_F(y, \delta) \subseteq \mathcal{I}_G(y, \delta + B), \qquad \mathcal{I}_G(y, \delta) \subseteq \mathcal{I}_F(y, \delta + B)}.$$

Proof. For $\theta \in \mathcal{I}_F(y, \delta)$, the triangle inequality gives

$$d_Y(G(\theta), y) \le d_Y(G(\theta), F(\theta)) + d_Y(F(\theta), y) \le B + \delta.$$

Interchanging F and G proves the other inclusion. □

This is BSC-CHN-03. It supplies no parameter-space Hausdorff stability or small parameter error without inverse regularity. A target must be constant, or satisfy its declared enclosing-radius condition, on the applicable enlarged set. Sampling uncertainty is added to δ under a separate coverage statement.

Proposition 14.12 (Marginal agreement and conditioned amplification)**.** *Let $|\omega_1\rangle, |\omega_2\rangle$ be orthogonal modes and*

$$|\psi\rangle = \frac{|\omega_1\rangle + |\omega_2\rangle}{\sqrt{2}}, \qquad |\phi\rangle = \frac{|\omega_1\rangle - |\omega_2\rangle}{\sqrt{2}}.$$

Their spectral-intensity laws are identical, hence their intensity-overlap score is 1*, but*

$$\langle \psi \mid \phi \rangle = 0, \qquad D_{\mathrm{tr}}(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = 1.$$

Moreover, let

$$\rho = \tfrac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|), \qquad \sigma = \tfrac{1}{2}(|0\rangle\langle 0| + |2\rangle\langle 2|).$$

Although $D_{\mathrm{tr}}(\rho, \sigma) = 1/2$, filtering onto $\mathrm{span}\{|1\rangle, |2\rangle\}$ *and normalizing on success yields orthogonal conditional states and distance* 1*.*

Proof. Both assertions follow by direct evaluation in the displayed orthonormal bases. The complete filter instrument including failure remains CPTP; only the normalized successful branch amplifies the distance. □

The first counterexample is BSC-QPH-02: equality of one measured marginal does not license equality or closeness of the full quantum state.

Proposition 14.13 (Driven open-system energy identity)**.** *Let a finite-dimensional differentiable open-system model satisfy*

$$\dot{\rho} = -\frac{i}{\hbar}[H(t), \rho] + \mathcal{D}_t(\rho), \qquad E(t) = \mathrm{Tr}[\rho(t)H(t)],$$

with a physical trace-preserving total generator and fixed energy zero. Then

$$\boxed{\dot{E} = \mathrm{Tr}(\rho\dot{H}) + \mathrm{Tr}(H\mathcal{D}_t(\rho))}. \tag{6}$$

Proof. Differentiate $\mathrm{Tr}(\rho H)$. Cyclicity gives $\mathrm{Tr}([H, \rho]H) = 0$, leaving (6). □

The two terms are conventionally drive/work power and dissipative energy flow under a fixed system–bath split. That split is not unique under strong coupling, feedback, Lamb-shift conventions or non-Markovian reduction; a time-dependent additive energy gauge changes the apparent work term. Dissipation can cool, heat or preserve energy. Measurements, resets, jumps, coupling energy and output fields require additional ledger terms. Thus a periodic drive can supply parametric gain without violating conservation, but (6) proves no experiment-specific gain. For a full bosonic cavity, direct use additionally requires a controlled finite-dimensional truncation or trace-class and domain hypotheses that justify differentiation and cyclicity. This is ledger claim BSC-ENE-01.

Theorem 14.14 (Scalar Bernoulli information and zero-error boundary)**.** *Let X be an input symbol and, conditional on $X = x$, suppose*

$$Y_1, \ldots, Y_N \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(q_x), \qquad K = \sum_{i=1}^{N} Y_i.$$

Then K is sufficient and

$$I(X; Y^N) = I(X; K) \le H(K) \le \log_2(N+1).$$

For uniform X on 256 symbols, $N \ge 255$ is necessary for exact decoding. It is not sufficient: zero-error recovery of 256 symbols is impossible for every finite N under this scalar conditionally-iid model.

Proof. For any binary string y of Hamming weight k,

$$\Pr(Y^N = y \mid X = x) = q_x^k (1 - q_x)^{N-k}.$$

The conditional likelihood depends on y only through K, and the conditional law of Y^N given K is independent of x. Since K is also a function of Y^N, the mutual informations agree, and K has at most $N + 1$ values. Eight bits therefore require $N + 1 \geq 256$. For finite N, every $q_x \in (0, 1)$ has full count support $\{0, \dots, N\}$, while $q_x = 0, 1$ supply only the two endpoint singleton supports. The 256 laws cannot be mutually singular, so proposition 6.4 blocks exact decoding. □

The sufficient-statistic statement is BSC-ENC-01 and the finite-sample zero-error obstruction is BSC-ENC-02.

The raw record contains N output bits per 8-bit input, so its nominal raw-output factor is $8/N$. Only $N < 8$ is compression by raw output-bit count. The wire cost is N bits only when every draw is transmitted; cached or receiver-local sampling has a different communications ledger. A sufficient count can be stored in $\lceil \log_2(N + 1) \rceil$ bits, but that compresses the repeated output, not the original symbol, and does not remove the zero-error obstruction. If M biases are independently queried n times, Hoeffding's inequality gives the standard learnability bound

$$\mathsf{P}\left\{\max_x |\widehat{q}_x - q_x| > \varepsilon\right\} \leq 2M \exp(-2n\varepsilon^2).$$

This is BSC-ENC-03. Query learnability is not a cryptographic security analysis.

Proposition 14.15 (Single-identity relation alignment)**.** *If square relation matrices C and S describe the same entity type in both argument positions, use the convention $P_{\phi(i),i} = 1$ for the permutation matrix of the identity bijection ϕ. Exact alignment then requires*

$$S = PCP^{\mathsf{T}}.$$

Independent row and column fits certify only $S = PCQ^{\mathsf{T}}$, which is strictly weaker in general.

Proof. Let $C = I$ and let $S = R \neq I$ be a permutation matrix. The independent choice $P = R, Q = I$ fits exactly, whereas $PIP^{\mathsf{T}} = I$ for every P. Thus no one-identity conjugation fit exists. □

Approximate one-identity alignment is generally a graph-matching or quadratic assignment problem. Two independent maps remain legitimate for separately typed row and column entities. Automorphisms can also leave the identity nonunique. This is BSC-SEM-01.

Finally, this report-channel form does not determine a fundamental coupling. For example, the low-energy electromagnetic fine-structure constant is

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}, \qquad \alpha^{-1} = 137.035\,999\,177(21)$$

in the 2022 CODATA adjustment [Mohr et al., 2025]. The operational-envelope axioms contain no equation fixing α and admit typed physical completions with different coupling values and, generally, different induced channels. Hence no universal numerical value follows from those axioms alone. Noninjectivity over one fixed report envelope would require a separately exhibited pair of operationally equivalent completions and is not asserted here. A derivation needs a typed electromagnetic action or Hamiltonian, normalization and renormalization conventions, a dimensionless prediction and metrological comparison. Recurrence of the integer 137 in a graph size, fit or scale is not such a bridge. This no-go is claim BSC-UNI-01.

Quantum networks and combs are prior art [Chiribella et al., 2009]. Dobrushin contraction, trace-norm contractivity, sufficient statistics, concentration bounds, open-system energy bookkeeping, and quadratic-assignment hardness are separately established results [Dobrushin, 1956, Pérez-García et al., 2006, Fisher, 1922, Hoeffding, 1963, Alicki, 1979, Sahni and Gonzalez, 1976]. The BSC contribution is their claim-relative integration with implemented reachable sets, evidence identity, demotion and explicit non-transfer to the full quantum morphism.

14.3 Electromagnetic evidence bridge

This section gives a typed electromagnetic completion of the operational channel core. Its ingredients are standard results in gauge theory, electromagnetism, network theory, metrology and renormalized quantum electrodynamics. The BSC contribution is only their placement on explicit evidence boundaries. In particular, the bridge sharpens BSC-UNI-01; it does not derive a value of α, identify a microscopic ontology, or propose a unified theory.

BSC-EM-01—Gauge descent. Let $\mathcal{A}(P)$ be the space of connections on a principal bundle P and let $\mathcal{G}(P)$ act by gauge transformations. A report $R : \mathcal{A}(P) \to \mathcal{Y}$ is physically gauge invariant exactly when it is constant on gauge orbits. Equivalently, there is a unique map $\overline{R}$ such that

$$R = \overline{R} \circ \pi, \qquad \pi : \mathcal{A}(P) \longrightarrow \mathcal{A}(P)/\mathcal{G}(P).$$

This is the universal property of the quotient. Curvature and closed-loop Wilson holonomy descend in this sense (for a non-Abelian connection, the holonomy conjugacy class or a traced Wilson loop descends). An open-path phase does not descend without typed endpoint states or a boundary trivialization.

BSC-EM-02—Curvature does not determine global holonomy. On $S^1 \times N$, let ϕ be the angular coordinate and take the dimensionless Abelian connections

$$a_\vartheta = \vartheta \, d\phi.$$

Every member has $da_\vartheta = 0$, whereas its loop holonomy is $\exp(2\pi i\vartheta)$. Integer shifts of ϑ are gauge equivalent, but distinct classes in $\mathbb{R}/\mathbb{Z}$ have the same curvature and different holonomy. Thus local field-strength evidence cannot by itself support a global phase claim. The physical observability of this distinction is the content of the Aharonov–Bohm effect [Aharonov and Bohm, 1959].

BSC-EM-03—Source compatibility. In differential-form notation, Maxwell's equations include

$$d\mathcal{F} = 0, \qquad d\mathcal{G} = \mathcal{J}.$$

Consequently,

$$d\mathcal{J} = d^2\mathcal{G} = 0.$$

Charge conservation is therefore a necessary compatibility condition for a globally smooth excitation $\mathcal{G}$; where such a global $\mathcal{G}$ exists, $\mathcal{J}$ is also exact. This implication does not reconstruct $\mathcal{G}$, fix constitutive data, or remove topological sectors. Singular sources, material interfaces and open boundaries require the corresponding distributional or relative-cohomology statement.

BSC-EM-04—Energy balance. For time-independent, symmetric, positive constitutive tensors $\mathbf{D} = \boldsymbol{\varepsilon}\mathbf{E}$ and $\mathbf{B} = \boldsymbol{\mu}\mathbf{H}$, define

$$u = \frac{1}{2}(\mathbf{E}\cdot\mathbf{D} + \mathbf{H}\cdot\mathbf{B}), \qquad \mathbf{S} = \mathbf{E}\times\mathbf{H}.$$

The curl equations and $\nabla\cdot(\mathbf{E}\times\mathbf{H}) = \mathbf{H}\cdot(\nabla\times\mathbf{E}) - \mathbf{E}\cdot(\nabla\times\mathbf{H})$ give Poynting's theorem [Poynting, 1884],

$$\partial_t u + \nabla\cdot\mathbf{S} = -\mathbf{J}\cdot\mathbf{E},$$

and hence, for a fixed region V,

$$\frac{d}{dt}\int_V u\,dV + \int_{\partial V}\mathbf{S}\cdot\mathbf{n}\,dA = -\int_V \mathbf{J}\cdot\mathbf{E}\,dV.$$

Temporal modulation, dispersion or active media require the omitted pump, material-state or storage channels to be restored before this balance is used as an audit equation.

BSC-EM-05—Passive scattering and hidden ports. For a linear time-invariant network with power-normalized propagating-port amplitudes $\mathbf{b} = S(\omega)\mathbf{a}$, no internal source and no release of initially stored energy, passivity gives

$$\|S\mathbf{a}\|_2^2 \le \|\mathbf{a}\|_2^2 \quad \text{for every } \mathbf{a} \quad \Longleftrightarrow \quad S^\dagger S \preceq I.$$

For non-power-normalized coordinates, the typed form is $S^\dagger W S \preceq W$ with the appropriate positive power metric W [Kurokawa, 1965, Marks and Williams, 1992]. If a complete lossless scattering operator is partitioned into observed and hidden ports,

$$U = \begin{pmatrix} S_{oo} & S_{oh} \\ S_{ho} & S_{hh} \end{pmatrix}, \qquad U^\dagger U = I,$$

then

$$I - S_{oo}^\dagger S_{oo} = S_{ho}^\dagger S_{ho}.$$

An observed power deficit can therefore represent unmeasured radiation, higher modes or other hidden ports as well as absorption. It does not type the loss mechanism by itself. Calibration uncertainty must also cross the boundary: if $\|\widehat{S} - S\|_2 \le \epsilon$, then

$$\sigma_{\max}(\widehat{S}) + \epsilon \le 1$$

certifies contraction, while $\sigma_{\max}(\widehat{S}) - \epsilon > 1$ falsifies the declared passive model. The interval between these tests is inconclusive.

BSC-EM-06—Magnitude does not determine phase. Under $\widehat{x}(\omega) = \int_{\mathbb{R}} x(t)e^{-i\omega t}\,dt$, take

$$S_0(\omega) = r, \qquad S_\tau(\omega) = re^{-i\omega\tau}, \qquad 0 < r \le 1, \quad \tau > 0.$$

Both are causal passive responses and satisfy $|S_0|^2 = |S_\tau|^2 = r^2$, but their phases, group delays and impulse responses $r\delta(t)$ and $r\delta(t-\tau)$ differ. Magnitude-only data therefore do not license phase, delay or time-waveform claims unless phase is measured or a separately verified theorem supplies conditions such as minimum phase and known delay.

Inverse claims are theorem-local. A boundary operator such as

$$\Lambda_{\varepsilon,\mu} : \mathrm{tr}_t\,\mathbf{E} \longmapsto \mathrm{tr}_t\,\mathbf{H}$$

has meaning only after the forward problem, coefficient class, geometry, gauge or diffeomorphism quotient, frequency set, trace spaces and full or partial data are fixed. A finite measured scattering matrix is not automatically the full boundary operator. Maxwell uniqueness results such as Kenig et al. [2011] establish conclusions under their stated geometric, regularity and data hypotheses; they do not provide a universal finite-band reconstruction of an arbitrary interior. At finite energy on a Lipschitz domain, tangential fields belong to the paired $H(\mathrm{curl})$ trace spaces rather than automatically being pointwise boundary vectors [Buffa et al., 2002].

BSC-EM-07—Field-normalization quotient. In rationalized units with $\hbar = c = 1$, consider

$$\mathcal{S}_{Z,\{q_i\},J} = -\frac{Z}{4}\int F_{\mu\nu}F^{\mu\nu}\, d^4x + \sum_i \mathcal{S}_i[\psi_i, d + iq_iA] - \int J^{\mu}_{\text{ext}} A_\mu \, d^4x.$$

Each matter action uses the displayed covariant derivative, so the expression also covers interactions whose dependence on A is not merely linear. For any $\lambda > 0$, the change $A' = \lambda A$ rewrites the same action with

$$Z' = \frac{Z}{\lambda^2}, \qquad q_i' = \frac{q_i}{\lambda}, \qquad J'_{\text{ext}} = \frac{J_{\text{ext}}}{\lambda}.$$

Thus the normalization-invariant couplings are

$$g_i = \frac{q_i}{\sqrt{Z}}, \qquad \alpha_i = \frac{q_i^2}{4\pi Z},$$

together with charge ratios. Substitution proves the invariance directly. Canonical field normalization sets a coordinate convention; gauge invariance and that convention do not select a numerical α_i. If a primitive charge q_0 and integers Q_i give $q_i = Q_i q_0$, then $\alpha_i = Q_i^2 \alpha_0$, so the lattice fixes ratios but still leaves one dimensionless boundary value.

BSC-EM-08—Flux-product quantization. For a species whose quantum phase is defined by the dimensionless connection $\mathcal{A}_i = (q_i/\hbar)A_{\text{phys}}$, a closed two-cycle Σ satisfies the first-Chern integrality condition

$$\frac{1}{2\pi}\int_\Sigma d\mathcal{A}_i = \frac{q_i}{2\pi\hbar}\int_\Sigma F_{\text{phys}} \in \mathbb{Z}.$$

This is a quantization of a charge–flux product, or equivalently of normalized holonomy. It does not determine q_i, Z, or $\alpha_i = q_i^2/(4\pi Z)$ separately. Dirac's monopole argument is the canonical physical instance [Dirac, 1931].

BSC-EM-09—Revised-SI metrology. At zero momentum transfer,

$$\alpha(0) = \frac{e^2}{4\pi\varepsilon_0\hbar c} = \frac{\mu_0 c e^2}{2h} = \frac{Z_0}{2R_{\mathrm{K}}}, \qquad R_{\mathrm{K}} = \frac{h}{e^2}.$$

In the revised SI, the numerical values of h, e and c are fixed, and the exact relation

$$\mu_0 = \frac{2h\,\alpha(0)}{ce^2}$$

means that μ_0 inherits the experimental uncertainty of $\alpha(0)$; the former exact value $4\pi \times 10^{-7}\,\mathrm{N\,A^{-2}}$ cannot be reused as independent evidence for α [Bureau International des Poids et Mesures, 2019, Mohr et al., 2025]. A genuine measurement bridge must name the observable and theory map. Atom-recoil determinations use, for an atom X,

$$\alpha^2 = \frac{2R_\infty}{c}\,\frac{A_{\mathrm{r}}(X)}{A_{\mathrm{r}}(e)}\,\frac{h}{m_X},$$

as in Morel et al. [2020]. An electron-magnetic-moment determination instead inverts a QED prediction for the measured anomaly [Fan et al., 2023]; a value obtained that way cannot then be counted as an independent test of the same theoretical terms. The 2026 atomic-hydrogen comparison likewise uses independently determined α and mass ratios as inputs to its bound-state-QED prediction; it is a stringent theory test, not an independent derivation of α [Maisenbacher et al., 2026].

BSC-EM-10—Renormalization requires a boundary value. The on-shell coupling is $\alpha_{\mathrm{OS}} \equiv \alpha(0)$, while a scheme-dependent running coupling may be written

$$\overline{\alpha}(\mu) = \frac{\overline{e}^2(\mu)}{4\pi}, \qquad \mu \frac{d\overline{e}}{d\mu} = \beta(\overline{e}).$$

The Ward–Takahashi identity constrains charge, vertex and field renormalization factors, but it does not supply their numerical boundary value [Ward, 1950, Takahashi, 1957]. Likewise, a renormalization-group equation transports a supplied value between scales; it does not create that value [Gell-Mann and Low, 1954]. Every running-coupling claim must therefore name the scheme, scale, threshold matching and observable. Atomic calculations normally use $\alpha(0)$ together with explicit radiative corrections; replacing it naively by a running $\alpha(\omega)$ can double count those corrections.

Binary-width and constructibility screen. The nearby integer patterns are exact but do not supply the missing physical map:

$$2^{31} - 1 = 2\,147\,483\,647, \qquad 2^{32} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65\,537,$$
$$65\,537 = 2^{16} + 1.$$

The first is the legacy signed-32-bit timestamp ceiling behind the Year 2038 problem [The Open Group, 2024]; the latter factorization by distinct Fermat primes explains the associated constructible regular polygons. For the prime 137,

$$\mathrm{ord}_{137}(2) = 68, \qquad 2^{31} - 1 \equiv 16, \quad 2^{32} - 1 \equiv 33, \quad 65\,537 \equiv 51 \pmod{137}.$$

Thus the binary expansion of the rational $1/137$ has period 68, but none of the displayed cutoff integers is divisible by 137. Moreover, the regular 137-gon is not straightedge-and-compass constructible because $(137-1)/2 = 68$ is not a power of two, unlike $(65\,537-1)/2 = 2^{15}$ [Wantzel, 1837]. These are arithmetic and representation facts. They neither identify the measured $\alpha^{-1} = 137.035\,999\,177(21)$ with 137 nor define a map to the invariant coupling $q_i^2/(4\pi Z)$.

BSC-EM-11—Aperiodic geometry-to-field descent. The Hat is an aperiodic monotile made from eight kites of the [3.4.6.4] Laves tiling, the deltoidal-trihexagonal dual of the (3.4.6.4) rhombitrihexagonal tiling [Smith et al., 2024a]. The periodic carrier kite grid and its aperiodic partition into Hats are distinct structures. Every Hat tiling mixes reflected and unreflected tiles; the later Spectre construction instead gives strictly chiral aperiodic monotiles [Smith et al., 2024b]. These are theorems about planar tilings and admitted isometries, not Maxwell spectra.

Let P be such a tiling partition and $\kappa = (\varepsilon, \mu, \sigma, \dots)$ a physical coefficient field. A translation-faithful materialization satisfies

$$\tau_v \kappa = \kappa \Longrightarrow \tau_v P = P, \qquad \mathrm{Stab}_{\mathrm{tr}}(\kappa) \subseteq \mathrm{Stab}_{\mathrm{tr}}(P).$$

Thus aperiodicity transfers to κ under this explicit hypothesis. Without it, constant coefficient fields and lattice-periodic coefficient fields materializing only the unlabeled carrier grid are exact counterexamples. Even faithful transfer establishes no band gap, localization, topological phase, optical activity, nonreciprocity, or device performance. A physical claim must bind

$$P \xrightarrow{C_{\mathrm{sel}}} X_N \xrightarrow{\Phi_{\mathrm{mat}}} (\varepsilon, \mu, \sigma, \Omega, \mathcal{B}) \xrightarrow{\mathcal{P}_{\lambda,p}} Y,$$

including the finite approximant, selector, scale, thickness, dispersion, loss, boundaries, illumination, polarization, calibration, and report.

For identical point scatterers $X_N = \{(x_j, y_j)\}_{j=1}^N$, the conditional scalar model is

$$F_N(k_x, k_y) = \sum_{j=1}^{N} e^{2\pi i(k_x x_j + k_y y_j)}, \qquad I_N = |F_N|^2.$$

It is not a full-wave solution. Indeed, point diffraction from the aperiodic Hat-vertex set can be two-periodic because of its underlying hexagonal net, whereas Spectre vertex and centroid diffraction is nonperiodic with chiral sixfold point symmetry [Kaplan et al., 2024a,b]. Real-space aperiodicity therefore neither implies nor contradicts reciprocal-space periodicity without a selector and observation model.

Moritake *et al.* selected Hat centroids, fabricated an H_6 approximant as 372 100 holes in a 350 nm SiN film, and observed sharp Bragg peaks whose positions were insensitive to illumination position, mirror-reversing pinwheel diffraction, and helicity-dependent intensity [Moritake et al., 2026]. For their declared golden-ratio/Fibonacci inflation geometry,

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad \theta_{\mathrm{chiral}} = \arccos\left(\frac{3\phi - 1}{4}\right) \approx 15.52^\circ.$$

This instantiates a specific geometry-to-fabricated-sample-to-report chain; it does not close the full Maxwell, calibration, or uncertainty obligations and does not prove the electromagnetic response of every Hat realization. The source-local observation record is `BSC-EM-OBS-01`; it is not the proof authority for BSC-EM-11. The eight kites, golden-ratio/Fibonacci inflation geometry, and sixfold intensity symmetry also supply no map to $q_i^2/(4\pi Z)$, no RG boundary value, and no derivation of $1/137$.

14.4 Corpus inheritance and local demotion

The prior reconstruction treated a hash-identified internal corpus as evidence under convention 1.3. Those source files are not independently inspectable in this public release, so the present reader cannot replay the textual comparison. Table 3 records this manuscript's source-local disposition of material not already represented by the transcribed tuples and morphism lineage. The words "retained" and "demoted" apply locally to the listed statement, not to an entire volume, and do not assert an independently verified quotation.

Table 3: Inheritance from the supplied BSC corpus.

Source	Retained mathematical content	Status boundary
Volume IV [Open Research Program, 2026a]	Identical observation laws imply a no-estimator obstruction; Fisher information additivity and data processing, rollout/cascade bounds, contraction criteria, entropy bookkeeping, and $I_{\text{classical}} \leq I_{\text{quantum}}$ remain scoped mathematical templates. A boundary Poynting checksum must use an absolute residual or an interval enclosure.	The proposed optical and computational predictions are unexecuted here. Signed cancellation is not accepted as a conservation certificate.
Volumes V–VI [Open Research Program, 2026b,c]	Exact quotient descent, non-descent of noninvariant queries, and Blackwell comparison are retained where their hypotheses match this paper.	Composite human-grounding scores, scaffold claims, and empirical programs are unexecuted and untested; humane intent is not a mathematical bridge.
Second Catalogue and Volume VII [Tree, 2026a, Open Boundary-State Research Program, 2026]	Explicit repair discipline, bounded claims, and dependency-aware research questions inform the certificate grammar.	Civic, historical, and long-future meditations are outside the mathematical dependency graph and receive no physical promotion.
Arithmetic Recursive Folds [Tree, 2026b]	Finite sieve identities and the declared Mellin-transform backbone survive as ordinary arithmetic analysis.	Primitive-orbit language is an analogy unless an operator and trace formula are constructed; no Riemann-hypothesis proof is inherited.
Certified Arithmetic Trace Reconstruction [Tree, 2026c]	The finite-matrix prime-comb obstruction and the locally finite arithmetic-density construction are retained in their stated classes.	The prime-local operator fold is conjectural; finite truncations do not certify an infinite spectral correspondence.
Derived Witnessed Descent [J. Tree research program, 2026]	Strict, derived, and observed-derived holonomy are separate gates; a locally finite 0-chain has a natural atomic Radon-measure realization; the stated distributionally closed smooth counterterm class collapses.	No derived or observed gate is promoted by satisfying only the strict gate.
Formal Verification supplement [Unattributed, 2026]	The bounded-jet orthogonal-prime-block obstruction is retained only under its explicit hypotheses and quantifiers.	Mechanical runs described in the source are source-asserted, not replayed: the generating scripts and accepted JSON/proof receipts were not supplied. Their computational status is therefore unexecuted here.

14.5 Explicit demotion rules

1. An undefined domain, codomain, unit, quotient, or boundary condition makes the claim-local verdict ill-posed and mathematical support none.
2. A counterexample makes the quantified mathematical verdict false; a repaired restricted claim receives a new identifier.
3. A proof sketch with an open lemma leaves the verdict open and supplies, at most, conditional or conjectural mathematical support.

4. Failure of a leakage gate invalidates promotion from the affected evaluation.
5. A numerical result without a reproducible receipt is unexecuted; a failed build is failed.
6. A verified preprint is not recorded as peer reviewed.
7. If an upstream dependency loses readiness, descendants receive only the declared readiness caps under definition 2.2; their verdicts do not propagate.
8. A proved upper enclosure below every prefix tolerance certifies the relevant coordinate. A straddling enclosure blocks certification but does not refute persistence. An actual discrepancy, certified lower bound, or qualified measurement above tolerance refutes persistence through that horizon; later recovery does not erase the excursion.
9. A successful transfer outside its declared domain is local only; using it in a new domain requires a new morphism certificate.
10. An analogy may motivate a test but cannot occupy a mechanism dependency slot.

15 Proven formal consequences

This section collects consequences established in the present paper. Their scope is exactly the hypotheses cited; no empirical coordinate is promoted.

Theorem 15.1 (Compatible-chain bounds)**.** *For a compatible chain of stochastic BSC morphisms $\mathfrak{M}_{0,1}, \ldots, \mathfrak{M}_{n-1,n}$, let η_i be the observation-channel contraction of the i-th post-processor and $\theta_i = \Theta_{i,i+1}$. Then*

$$\Theta_{0n} \leq \theta_{n-1} + \sum_{i=0}^{n-2} \theta_i \prod_{j=i+1}^{n-1} \eta_j.$$

For a common parameter family,

$$\delta(\mathsf{E}_0, \mathsf{E}_n) \leq \sum_{i=0}^{n-1} \delta(\mathsf{E}_i, \mathsf{E}_{i+1}).$$

Proof. Iterate theorems 7.6 and 7.7. □

Corollary 15.2 (Persistence obstruction)**.** *If, at any prefix in a fold chain, the actual identity, viability, naturality, residual, or deficiency coordinate—or a certified lower bound for it—exceeds its declared tolerance, persistent identity through that prefix is false. If only a certified upper bound exceeds tolerance, the chain fails to certify persistence but does not by that fact refute it.*

Proof. This is the conjunction of definitions 7.2 and 9.2 and theorem 9.3; the distinct coordinates are not allowed to compensate one another. □

Theorem 15.3 (Operational factorization criterion)**.** *Let $q : X \to \bar{X}$ be an exact operational quotient. A channel $\Phi : X \rightsquigarrow Z$ induces a unique channel $\bar{\Phi} : \bar{X} \rightsquigarrow Z$ if and only if Φ is constant on the fibers of q, under the standard Borel hypotheses of theorem 6.2.*

Proof. The forward implication follows from $\Phi = \bar{\Phi} \odot \delta_q$; the reverse implication is theorem 6.2. □

Corollary 15.4 (Discard obstruction)**.** *If two joint states have the same declared reduced quotient but different transformed accessible probabilities, no transform channel descends to that reduced quotient.*

Proof. Contrapositive of theorem 15.3. □

Theorem 15.5 (DAG readiness-cap monotonicity)**.** *On a finite claim DAG, the recursion in definition 2.2 determines a unique greatest readiness assignment below the initial readiness and every declared cap. Readiness can only remain fixed or decrease unless initial evidence, edges, or cap maps change. Every claim-local mathematical verdict is unchanged by dependency propagation.*

Proof. Each readiness axis is a finite meet-semilattice. Process vertices in a topological order. At each vertex, all predecessor values are already fixed, and meeting the initial value with the monotone cap values gives the greatest element satisfying those inequalities. Induction over the topological order gives existence, uniqueness, and greatestness. The recursion has no verdict update, so verdicts remain claim-local. □

Corollary 15.6 (Topology-to-charge block)**.** *A computed topological class does not promote a physical-charge claim unless a certified bridge χ is present as a dependency.*

Proof. Apply proposition 13.3 and theorem 15.5. □

16 Worked fixtures and counterexamples

Each fixture is intentionally small. It fixes inputs and expected status so that future implementations can be tested against a stable result.

Worked fixture 16.1 (Torus winding without electric charge)**. Inputs and types.** Let $T^2 = S^1 \times S^1$ and

$$\gamma : [0, 2\pi] \to T^2, \qquad \gamma(t) = (e^{2it}, e^{-3it}).$$

The admissible observable is the pair of winding integrals

$$w_j(\gamma) = \frac{1}{2\pi i} \int_{\gamma_j} z^{-1}\, dz \in \mathbb{Z}.$$

Assumptions. Endpoints are identified, orientation is increasing t, and homology has coefficients $\mathbb{Z}$. **Claimed result.**

$$[\gamma] = (2, -3) \in H_1(T^2; \mathbb{Z}) \cong \mathbb{Z}^2.$$

Test. Substitute $z_1 = e^{2it}$, $z_2 = e^{-3it}$ in the integrals, or reduce the lifted endpoint increments by 2π. **Expected certificate.** Mathematical status proved; source present calculation; computation not required; transfer to electric charge blocked. **Failure and demotion.** Any output q_{electric} without a typed $\chi : \mathbb{Z}^2 \to \mathbb{R}\,\mathrm{C}$ fails definition 13.2. **Evidence that changes the result.** A gauge/action/current or calibrated measurement derivation of χ, with normalization, units, matter representation, and anomaly checks, can open the physical-charge dependency; it cannot change the homology result.

Worked fixture 16.2 (One-dimensional Dirichlet-to-Neumann ambiguity)**. Inputs and types.** Let $\gamma \in L^\infty(0,1)$ satisfy $0 < \gamma_- \le \gamma \le \gamma_+ < \infty$, and solve

$$-(\gamma u')' = 0, \qquad u(0) = f_0, \quad u(1) = f_1.$$

The trace is $\tau_\partial u = (u(0), u(1)) \in \mathbb{R}^2$; extension sends $f \in \mathbb{R}^2$ to $u_f \in H^1(0,1)$; the response is $\Lambda_\gamma : \mathbb{R}^2 \to \mathbb{R}^2$ and sends f to outward flux. Put $R_\gamma = \int_0^1 \gamma(x)^{-1}\, dx$. **Claimed result.**

$$\Lambda_\gamma = \frac{1}{R_\gamma} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The distinct conductivities

$$\gamma_1 \equiv 1, \qquad \gamma_2(x) = \begin{cases} 2, & 0 < x < 1/2, \\ 2/3, & 1/2 < x < 1 \end{cases}$$

both have $R_\gamma = 1$, hence identical response. **Test.** Constant flux gives $\gamma u' = (f_1 - f_0)/R_\gamma$; apply outward signs at the endpoints. **Noise bound.** In Euclidean operator norm,

$$\|\Lambda_R - \Lambda_{\widetilde{R}}\|_2 = 2|R^{-1} - \widetilde{R}^{-1}|.$$

For $v = (1, -1)^\top$, define the estimator

$$\widehat{g} = \tfrac{1}{4} v^\top \widehat{\Lambda} v.$$

Then $\|\widehat{\Lambda} - \Lambda_R\|_2 \le \eta$ implies

$$|\widehat{g} - R^{-1}| \le \tfrac{1}{4}\|v\|_2^2 \eta = \eta/2.$$

Thus the data identify only effective conductance $g = R^{-1}$ to this error, not the spatial profile. **Expected certificate.** Forward solution and DN formula proved; profile reconstruction non-identifiable; noise statement bounded. **Failure and demotion.** "The boundary reconstructs $\gamma(x)$" is refuted for this class. **Evidence that changes the result.** A restricted parametric class, interior data, additional frequencies, or a proved injective data map with stability can yield a new scoped claim.

Worked fixture 16.3 (Exact Blackwell order and directed deficiency)**. Inputs and types.** Let $\Theta = \{0, 1\}$ and $Y_\ell = Y_m = \{0, 1\}$. The source experiment is perfect:

$$P_{\ell,0} = (1, 0), \qquad P_{\ell,1} = (0, 1).$$

The target is a binary symmetric garbling:

$$P_{m,0} = (3/4, 1/4), \qquad P_{m,1} = (1/4, 3/4).$$

Assumptions. Total variation is half the ℓ^1 distance; all Markov matrices are allowed. **Claimed result.**

$$\delta(\mathsf{E}_\ell, \mathsf{E}_m) = 0, \qquad \delta(\mathsf{E}_m, \mathsf{E}_\ell) = 1/4, \qquad \Delta(\mathsf{E}_\ell, \mathsf{E}_m) = 1/4.$$

Test. The $1/4$-crossover channel maps E_ℓ exactly to E_m. Conversely let r_y be the probability of output 1 after observing y under a proposed reverse channel. Its two probabilities are $p_0 = (3r_0 + r_1)/4$ and $p_1 = (r_0 + 3r_1)/4$, so $p_1 - p_0 = (r_1 - r_0)/2 \le 1/2$. Errors below $1/4$ would require $p_0 < 1/4$ and $p_1 > 3/4$, a contradiction. The identity reverse channel attains $1/4$. **Expected certificate.** Exact rational proof; mathematical verdict true with proved support; source and target direction explicit. **Failure and demotion.** Reversing the prose or calling directed deficiency symmetric is a type/status failure. **Evidence that changes the result.** Enlarging the observation or restricting the parameter/decision class defines a new experiment and may change the deficiency.

Worked fixture 16.4 (Koopman pseudomode and spectral pollution)**. Inputs and types.** Let $X = \{-1, 1\}^{\mathbb{Z}}$ with product measure and the two-sided shift $F(x)_j = x_{j+1}$. Put $\phi_j(x) = x_j$, so $\mathcal{K}_F \phi_j = \phi_{j+1}$ and the ϕ_j are orthonormal in $L^2(\mu)$. For $|\lambda| = 1$,

$$\psi_N = N^{-1/2} \sum_{j=0}^{N-1} \lambda^{-j} \phi_j.$$

Claimed result.

$$\|\mathcal{K}_F\psi_N - \lambda\psi_N\|_2 = \sqrt{2/N}.$$

Yet compression to $\mathcal{G}_N = \operatorname{span}\{\phi_0, \dots, \phi_{N-1}\}$ is a nilpotent unilateral truncation with every eigenvalue 0, whereas the true Koopman operator is unitary and $0 \notin \sigma(\mathcal{K}_F)$. **Test.** Interior coefficients telescope; only orthogonal endpoint terms $-\lambda\phi_0/\sqrt{N}$ and $\lambda^{-(N-1)}\phi_N/\sqrt{N}$ remain. Direct matrix compression gives a nilpotent shift. **Expected certificate.** Symbolic residual proved; the zero eigenvalues are certified spectral pollution; no empirical execution claimed. **Failure and demotion.** Interpreting ψ_N as an exact eigenmode, or compression eigenvalue 0 as physical decay, is refuted. **Evidence that changes the result.** A residual-controlled limit in a declared topology, perturbation stability, and an applicable Koopman learnability theorem can promote a specified spectral claim.

Worked fixture 16.5 (Finite $\mathbb{Z}_2$ quantum-reference-frame descent)**. Inputs and types.** Let $G = \mathbb{Z}_2$, $\mathcal{H}_R = \ell^2(G)$, and $\mathcal{H}_S = \mathbb{C}^2$. The frame carries the left-regular action $L_h|g\rangle = |h+g\rangle$, and the system carries $U_S(h) = X^h$. Define the frame-trivialization unitary

$$W = \sum_{g\in G} |g\rangle\langle g|_R \otimes X_S^g.$$

Write $\mathrm{Ad}_W(\rho) = W\rho W^\dagger$. The target accessible algebra is $I_R \otimes \mathcal{B}(\mathcal{H}_S)$. Its paired pullback is the relational algebra

$$W^\dagger(I_R \otimes A)W = \sum_{g\in G} |g\rangle\langle g|_R \otimes X^g A X^g,$$

which is invariant under the simultaneous action $L_h \otimes X^h$. Let

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad |\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

and $\rho_\Phi = |\Phi^+\rangle\langle\Phi^+|$, $\rho_\Psi = |\Psi^+\rangle\langle\Psi^+|$. Both states are invariant under $X_R \otimes X_S$, and both have the same reduced system state $I_S/2$. **Claimed result.**

$$\mathrm{Tr}_R(W\rho_\Phi W^\dagger) = |0\rangle\langle 0|, \qquad \mathrm{Tr}_R(W\rho_\Psi W^\dagger) = |1\rangle\langle 1|.$$

Their trace distance is 1; the expectations of Z_S are $+1$ and -1. Hence no channel on the common reduced input $I_S/2$ can factor $\mathrm{Tr}_R \circ \mathrm{Ad}_W$ through Tr_R on a class containing ρ_Φ, ρ_Ψ. **Test.** Here W is $\mathrm{CNOT}_{R\to S}$, and exact ket calculation gives

$$W|\Phi^+\rangle = |+\rangle_R|0\rangle_S, \qquad W|\Psi^+\rangle = |+\rangle_R|1\rangle_S.$$

Equivalently, for the discard–reset–transform route with $\iota_0(\rho_S) = |0\rangle\langle 0|_R \otimes \rho_S$,

$$\mathrm{Tr}_R\,\mathrm{Ad}_W(\iota_0(I_S/2)) = I_S/2,$$

which agrees with neither transform–discard output. **Expected certificate.** Exact 4×4 matrix or ket calculation; invariance of both inputs, paired relational algebra, and nonfactorization proved on the declared accessible algebra. **Failure and demotion.** No reduced transformed channel may be inferred from $\mathrm{Tr}_R\,\rho$ alone on a class containing joint correlations that alter the output. **Evidence that changes the result.** A restricted state class, final-frame localizability condition, or proved factorization through the quotient can make the operations agree. **Scope.** This is a finite-group QRF fixture, not a universal model of all quantum reference frames.

Worked fixture 16.6 (Locally satisfiable parity data that do not glue)**. Inputs and types.** Let vertex variables $x_i \in \mathbb{Z}_2$ on a 3-cycle and relation stalks

$$R_{12} : x_1 + x_2 = 0, \quad R_{23} : x_2 + x_3 = 0, \quad R_{31} : x_3 + x_1 = 1.$$

Restriction from an edge relation to either endpoint is coordinate projection. **Assumptions.** Addition is modulo 2. **Claimed result.** Every edge relation is nonempty and every endpoint projection is $\{0,1\}$, but the inverse limit of the three relations is empty. **Test.** Adding all equations gives $2(x_1 + x_2 + x_3) = 1$ in $\mathbb{Z}_2$, hence $0 = 1$. Equivalently the parity cocycle $(0,0,1)$ has cycle sum 1 and is the nonzero class in $H^1(C_3;\mathbb{Z}_2)$. **Expected certificate.** Exact finite contradiction and a declared coefficient complex; local validity preserved, global-section claim refuted. **Failure and demotion.** Individual plausibility or surjective overlap projections do not promote a global state. **Evidence that changes the result.** Altering one parity, adding a slack/noise model with a stated consistency radius, or identifying a faulty edge yields a different scoped gluing problem.

Worked fixture 16.7 (Off-shell massive-field shift)**. Inputs and types.** All variables in this fixture are nondimensionalized. On $\Omega = (0,1)$, put

$$H^2_N(0,1) = \{\phi \in H^2(0,1) : \phi'(0) = \phi'(1) = 0\}$$

and define

$$\mathcal{E}_m : H^2_N(0,1) \to L^2(0,1), \qquad \mathcal{E}_m(\phi) = -\phi'' + m^2\phi, \quad m > 0.$$

The source equation is $\mathcal{E}_m(\phi) = 0$. Let $T_c\phi = \phi + c$, $c \in \mathbb{R}$. **Claimed result.**

$$R_c(\phi) = \mathcal{E}_m(T_c\phi) - \mathcal{E}_m(\phi) = m^2 c, \qquad \|R_c\|_{L^2(0,1)} = m^2|c|.$$

Neumann boundary conditions are preserved, but the target field equation is not for $c \neq 0$. **Test.** Differentiate the constant and substitute. **Expected certificate.** Exact dimensionless residual, source equation and target equation stated; alleged symmetry blocked off shell. **Failure and demotion.** Geometric resemblance or boundary compatibility cannot compensate for nonzero equation residual. **Evidence that changes the result.** The cases $c = 0$, $m = 0$, or a modified action/source transformation define different claims; an on-shell map requires a proof for the full solution class.

Worked fixture 16.8 (Permanent exact regression counterexample)**. Inputs and types.** The quantified claim is $\forall x \in \mathbb{R}, \sqrt{x^2} = x$. The exact integer input is $x = -1$; the principal square root is computed as $\mathrm{isqrt}(x^2) = 1$. **Claimed result.** $1 \neq -1$, so the quantified claim is false; the correct real identity is $\sqrt{x^2} = |x|$. **Mechanical test.** The retained script `verify_counterexample.py`, run under CPython 3.12.13, emitted a deterministically serialized, key-sorted JSON receipt twice with byte-identical output. The fail-closed checker validates the shipped schema and independently recomputes $\mathrm{isqrt}(x^2)$; permanent negative tests reject a schema-invalid generator and a false arithmetic generator. The generator, schema, retained receipt, and checker SHA-256 values are respectively `f56d833f54b537d79111a08aa432e455c6e2b7223c44c3f536c73e4d9ea20dde`, `786edf0427324310048f1060ffab594d73aec137529d46aec14225d44b20c311`, `491eb5c6b4cec20b457010003197bd8472236810a28f304bef89cd7caf20cc72`, and `ad2c0f9797a37b4ae56f48d45727961565e478b19e7f149a768190d2c5c28eae`. **Expected certificate.** Mathematical verdict false with proved support for the universal claim; computational status exact receipt; no floating arithmetic. **Failure and demotion.** Any future rewrite that returns the original universal identity passes formatting but fails the permanent regression. **Evidence that changes the result.** Restricting $x \geq 0$ creates a true new proposition; it does not repair the original quantified claim.

Worked fixture 16.9 (Documentary zeta–DQPT scope fixture)**. Inputs and types.** For $N \in \mathbb{N}$, $\beta_{\mathrm{eff}} > 0$, $t \in \mathbb{R}$, and $s = \beta_{\mathrm{eff}} + it$, put

$$Z_N(\beta_{\mathrm{eff}}) = \sum_{n=1}^{N} n^{-\beta_{\mathrm{eff}}}, \qquad S_N(s) = \sum_{n=1}^{N} (-1)^{n+1} n^{-s},$$

and define the ideal accumulated phase factor

$$\mathcal{L}_N(\beta_{\text{eff}}, t) = -\frac{S_N(s)}{Z_N(\beta_{\text{eff}})}.$$

The source is the journal article and its supplementary material pinned in section 17.4; no experimental data or control program is an input to this fixture. **Claimed result.**

$$Z_N(\beta_{\text{eff}})\mathcal{L}_N(\beta_{\text{eff}}, t) + S_N(s) = 0$$

exactly. **Test.** Substitute the definitions for the finite identity. **Expected certificate.** BSC-ZDQ-01 is true with proved support. F9 is documentary and computationally unexecuted: it contains no checker, receipt, BSC replay of the reported NMR experiment, or high-zero simulation. The general normalized-scale, singularity-support, local-root, and contour theorems proved earlier in the framework, together with the zeta-specific hypotheses in section 17.4, support BSC-ZDQ-02a–02e. This fixture executes neither that general mathematics nor the application-specific transfer. **Failure and demotion.** A finite-resolution observation $|\widehat{\mathcal{L}}_N| \leq \varepsilon$ does not by itself certify $\zeta(s) = 0$, a complete zero count, or the Riemann Hypothesis. Failure of an implementation residual demotes the implementation edge, not the exact finite identity. **Evidence that changes the result.** An exact checker and retained receipt can promote this fixture's computational axis. Guarded complex enclosures and a complete zero-count calculation belong to a separate application certificate. Public raw data, pulse programs, calibration and an independent replication can promote the empirical axes.

Worked fixture 16.10 (Executable coupled-surrogate host dependence)**. Inputs and types.** Use exact rational arithmetic, $z_k = 0$, $\widehat{z}_k = 1/100$, $E_0^{(h)} = 0$, horizon $H = 10$, tolerance $\tau = 1/20$, and

$$E_{k+1}^{(h)} = a_h E_k^{(h)} + \frac{1}{100}.$$

The two stable hosts have $a_A = 1/2$ and $a_B = 9/10$. **Claimed result.** Both hosts receive exactly the same standalone interface error, but

$$E_{10}^{(A)} = \frac{1023}{51200} < \tau, \qquad E_{10}^{(B)} = \frac{6513215599}{100000000000} > \tau.$$

Host A remains within tolerance at every prefix. Host B first violates the tolerance at $k = 7$. **Mechanical test.** The retained receipt in `fixtures/F10_coupled_surrogate/` records every exact prefix. Its fail-closed checker independently recomputes the recurrence, validates the schema and factored identities, executes the generator in an isolated temporary directory, and requires byte-identical regeneration. Negative tests reject a stale host hash, an altered horizon with stale arithmetic, a false within-tolerance disposition for Host B, a decimal substituted for an exact rational, and attempted receipt overwrite. **Expected certificate.** The universal statement that equal standalone surrogate error entails equal coupled-host tolerance disposition is false, with proved support and an exact CPython 3.12.13 receipt. This one-coordinate result is not a certificate of full claim-relative admissibility. **Failure and demotion.** The fixture is code-verification evidence for this finite recurrence only. It supplies no empirical or physical validation and no guarantee for another host, operating domain, coupling, horizon, or simulator. **Evidence that changes the result.** A different host or horizon is a new claim identity and must be recomputed or covered by a certified compatibility map.

17 Application boundaries: holography, QCD, primordial-field theory, engineered arithmetic dynamics, and learned models

17.1 Holography and reconstruction

BSC can audit a holographic claim, but does not prove holography. A valid application must identify the bulk and boundary theories, state spaces and observable algebras, code subspace, reconstruction map, gauge quotient, semiclassical or large-parameter limit, error topology, and physical dictionary. An inverse-PDE uniqueness theorem and a holographic duality are different claims. The former cannot be promoted by the visual fact that data reside on a boundary; the latter cannot be promoted by the generic quotient theorems in this paper.

17.2 QCD and controlled confinement comparisons

The monopole–center-vortex constructions cited in section 13 provide a stronger comparator because their carrier and bridge are internal to a specified gauge theory. Even there, controlled compactified regimes, anomaly arguments, condensation criteria, and string tensions do not automatically supply physical $\mathbb{R}^4$ QCD, generations, electroweak charge, hadron spectra, or scattering. A BSC certificate must preserve that domain boundary.

17.3 Klingman's primordial-field papers

The primordial-field papers propose a candidate structure consisting of a self-interacting field, dual field components, toroidal configurations, complex or Kähler duality, closed paths and winding, stabilization through competing effects, and a winding-to-charge correspondence [Klingman, 2024, 2022]. A later paper proposes a qualitative solenoidal-flux account of confinement and expressly leaves a detailed quantitative treatment to future work [Klingman, 2025]. These are legitimate objects for formalization and testing. The current classification is:

Candidate grammar: topological carrier and stabilization model.
Not established: a derivation of quark charge, QCD, or hadron dynamics.

The papers' toroidal parametrization supplies a winding computation, but a parametric flow is not yet a solution generated by a well-typed physical action. The exact bridge still owed is displayed in table 4.

Failure of promotion does not erase useful mathematics: winding classes, stability conditions, and toroidal boundary-value problems can survive as mathematical candidate components. BSC does not validate primordial-field theory, and the theory does not validate BSC.

17.4 Riemann zeros and DQPTs: a normalized-scale instance

Riemann zeros and dynamical quantum phase transitions (DQPTs) are treated here as an instance of the general scale framework above. The source pinned here is the article by Wei et al. [2026], DOI 10.1038/s41467-026-74935-8. The publisher record was checked on 30 July 2026: it listed publication on 1 July 2026 and still identified the available text as an unedited early-access article in press. Later copy-editing, pagination, data release, or corrections are not silently inherited by this release record.

Mathematical provenance. The eta factorization, Abel summation, the Euler transformation of an alternating tail, harmonic-sum asymptotics, isolated holomorphic zeros, and Rouché's theorem are classical imported ingredients. The explicit bounds, normalization diagnosis,

Table 4: Bridge obligations for physical promotion of the primordial-field candidate.

Required object	Inspectable test	Current status
Action and field space	variational derivative; dimensions; boundedness or domain of validity	not supplied in a form deriving the claimed torus
Gauge covariance and conserved current	transformation law, Noether/Gauss identity, charge normalization	bridge χ absent
Boundary and regularity class	well-posed field equations and finite energy	not established for the full candidate
Anomaly cancellation or matching	fermion content and anomaly polynomial or controlled matching condition	not established
Spin and statistics	representation of the Lorentz/rotation group and quantization rule	not established
Color structure	$SU(3)$ representations, gauge connection, observables	not established
Confinement criterion	Wilson/'t Hooft behavior or gauge-invariant equivalent with regime	qualitative solenoidal analogy only; source report
Running and scale	renormalization prescription and beta function	not established
Generations and mixing	spectrum and mixing matrix from the action	not established
Spectra and scattering	hadron masses, amplitudes, uncertainties, and independent falsifiers	not established

pointwise rate split, singular-set and slice corollaries, and local root-displacement formula below are self-contained consequences assembled here from those ingredients. No historical-priority claim is made for them.

Exact finite algebra. For $s = \beta_{\text{eff}} + it$, define

$$H_{0,N} = \sum_{n=1}^{N} \log n\, |n\rangle\langle n|, \qquad Z_N(\beta_{\text{eff}}) = \sum_{n=1}^{N} n^{-\beta_{\text{eff}}},$$

$$\rho_{N,\beta_{\text{eff}}} = Z_N(\beta_{\text{eff}})^{-1} \sum_{n=1}^{N} n^{-\beta_{\text{eff}}} |n\rangle\langle n|,$$

$$P_N = \sum_{n=1}^{N} (-1)^n |n\rangle\langle n|, \qquad S_N(s) = \sum_{n=1}^{N} (-1)^{n+1} n^{-s}.$$

The ideal probe coherence is therefore

$$\mathcal{L}_N(\beta_{\text{eff}}, t) = \operatorname{Tr}\Big(\rho_{N,\beta_{\text{eff}}} e^{-itH_{0,N}} P_N\Big) = -\frac{S_N(s)}{Z_N(\beta_{\text{eff}})}. \tag{7}$$

This is an exact finite identity, not an analogy and not a thermodynamic statement. With

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-s} = (1 - 2^{1-s})\zeta(s), \qquad \Re s > 0,$$

it gives $Z_N \mathcal{L}_N = -S_N \to -\eta = (2^{1-s} - 1)\zeta(s)$ at fixed s. The normalization is not innocuous. For fixed s with $0 < \beta_{\text{eff}} \leq 1$, $Z_N(\beta_{\text{eff}}) \to \infty$ while $S_N(s) \to \eta(s)$, and therefore

$$\mathcal{L}_N(\beta_{\text{eff}}, t) \longrightarrow 0$$

whether or not $\zeta(s) = 0$. Raw limiting coherence is thus not a zero discriminator in the critical strip. An admissible descendant must use the renormalized amplitude $Z_N\mathcal{L}_N$, a finite-N root certificate, or a separately defined rate-function limit. This is proposition 10.4 with carrier $A_N = -S_N$ and positive normalizer Z_N: normalization preserves every finite zero but collapses the raw pointwise limit.

Proposition 17.1 (Certified alternating-tail transfer). *Let S_N and η be as above. If $\sigma = \Re s > 0$, then*

$$|\eta(s) - S_N(s)| \le N^{-\sigma}\left(1 + \frac{|s|}{\sigma}\right). \tag{8}$$

Consequently $S_N \to \eta$ locally uniformly on the half-plane $\Re s > 0$. More precisely, for compact $K \subset \{\Re s > 0\}$, with $\sigma_K = \min_{s\in K}\Re s$ and $M_K = \max_{s\in K}|s|$,

$$\sup_{s\in K}|\eta(s) - S_N(s)| \le N^{-\sigma_K}\left(1 + \frac{M_K}{\sigma_K}\right).$$

Proof. Let $a_n = (-1)^{n+1}$ and $A(x) = \sum_{n\le x} a_n$, so $|A(x)| \le 1$. Abel summation applied after N, followed by $\frac{d}{dx}x^{-s} = -sx^{-s-1}$, bounds the endpoint contribution by $N^{-\sigma}$ and the integral contribution by

$$|s|\int_N^\infty x^{-\sigma-1}\,dx = \frac{|s|}{\sigma}N^{-\sigma}.$$

This proves (8); taking the extrema on K proves the uniform statement. □

Proposition 17.2 (Fixed-s scaling discriminator). *Fix $s = \beta + it$ with $0 < \beta < 1$, write $\mathcal{L}_N(s) = \mathcal{L}_N(\beta, t)$, and define, for $N \ge 2$ whenever $\mathcal{L}_N(s) \ne 0$,*

$$\alpha_N(s) = -\frac{\log|\mathcal{L}_N(s)|}{\log N}.$$

Then $\mathcal{L}_N(s) \ne 0$ for all sufficiently large N, and

$$\lim_{N\to\infty}\alpha_N(s) = \begin{cases} 1-\beta, & \eta(s) \ne 0, \\ 1, & \eta(s) = 0. \end{cases}$$

Proof. Put $R_N(s) = \eta(s) - S_N(s)$, $m = N+1$, $f_n = n^{-s}$, and $\Delta f_n = f_n - f_{n+1}$. The one-step Euler identity is

$$\sum_{k=0}^\infty (-1)^k f_{m+k} = \frac{1}{2}f_m + \frac{1}{2}\sum_{k=0}^\infty (-1)^k \Delta f_{m+k}.$$

Since $\Delta f_n = O_s(n^{-\beta-1})$ and $\Delta^2 f_n = O_s(n^{-\beta-2})$, Abel summation bounds the second alternating series by

$$O_s\left(|\Delta f_m| + \sum_{n=m}^\infty |\Delta^2 f_n|\right) = O_s(m^{-\beta-1}).$$

Restoring the sign of the first omitted term gives

$$R_N(s) = (-1)^N\left(\frac{1}{2}(N+1)^{-s} + O_s(N^{-\beta-1})\right).$$

On every fixed compact $U \Subset \{\Re s > 0\}$, the difference estimates are uniform: if the displayed error is $E_N(s)$, then

$$|E_N(s)| \le C_U N^{-\Re s-1}, \qquad s \in U.$$

Integral comparison gives

$$Z_N(\beta) = \frac{N^{1-\beta}}{1-\beta} + O_\beta(1).$$

If $\eta(s) \neq 0$, equation (7) yields

$$\mathcal{L}_N(s) = -(1-\beta)\eta(s)N^{\beta-1}(1+o(1)).$$

If $\eta(s) = 0$, then $S_N = -R_N$, so

$$\mathcal{L}_N(s) = (-1)^N \frac{1-\beta}{2} N^{-1-it}(1+o(1)).$$

Both expansions are eventually nonzero, and taking logarithms proves the claim. □

In the notation of definition 10.3, this instance has

$$\lambda_N = \log N, \qquad \gamma(s) = 1-\beta, \qquad \kappa(s) = \begin{cases} 0, & \eta(s) \neq 0, \\ \beta, & \eta(s) = 0. \end{cases}$$

Theorem 10.5 separates a normalization background from the exceptional carrier exponent. Since $1 - 2^{1-s} \neq 0$ in the open critical strip, the second branch is exactly the zeta-zero branch. Thus raw coherence decays throughout the strip, but its fixed-s decay exponent changes at a zero. Promoting that exponent to a measured-zero claim still requires a finite-size error budget and implementation certificate.

For the source's qubit scaling $N = 2^d$, the declared finite-size free-energy rate is

$$\mathcal{F}_{1,N}(s) = -\frac{1}{d}\log|\mathcal{L}_N(s)| = (\log 2)\alpha_N(s).$$

Proposition 17.2 therefore proves the pointwise fixed-s formula

$$\mathcal{F}_1(s) = \begin{cases} (1-\beta)\log 2, & \eta(s) \neq 0, \\ \log 2, & \eta(s) = 0. \end{cases}$$

This is the mathematical rate-value bridge used in the source's DQPT interpretation. It is not uniform near a zero, does not interchange a growing-t limit with $N \to \infty$, and is not a finite-size experimental certificate.

Corollary 17.3 (Pointwise rate singularity set)**.** *On $0 < \Re s < 1$, the discontinuity set of $\mathcal{F}_1$ is exactly the zero set of ζ. At a zero $s_0 = \beta_0 + it_0$, its value exceeds the limiting off-zero background by $\beta_0 \log 2$. Consequently,*

$$\mathrm{RH} \iff \mathrm{Disc}(\mathcal{F}_1) \subseteq \{s : \Re s = 1/2\} \quad \textit{within } 0 < \Re s < 1.$$

Proof. Apply theorem 10.8 with $\Sigma = \{s : \eta(s) = 0\}$, $\gamma(s) = (1-\Re s)\log 2$, and $\rho(s) = (\Re s)\log 2$. The eta zero set is locally discrete, and its prefactor is nonzero in the open strip, so it is precisely the nontrivial zeta zero set. The final equivalence is therefore a restatement of the RH zero-location predicate. □

This corollary is a representation equivalence in the ideal model, not a decision of either side of RH and not an empirical zero certificate.

Corollary 17.4 (Fixed-β real-time slice)**.** *For fixed $0 < \beta < 1$, let $\iota_\beta(t) = \beta + it$. Then*

$$\mathrm{Disc}(\mathcal{F}_1 \circ \iota_\beta) = \{t \in \mathbb{R} : \zeta(\beta + it) = 0\},$$

and every such sliced jump has size $\beta \log 2$.

Proof. Proposition 10.9 gives the general inclusion. Here the formula is constant at $(1-\beta)\log 2$ off the isolated zeros on the line and equals $\log 2$ at each zero. Nonzero points on the line approach every such isolated zero, proving visibility and equality. $\square$

In the exact limiting model, zero membership also descends through the scalar rate by the Borel decision cell $\{\log 2\}$, because $\mathcal{F}_1(s) = \log 2$ exactly on the zero set. This is a statement about the exact pointwise limit. A finite noisy estimator does not inherit that decision without the separation required by proposition 6.4. Moreover, proposition 10.7 shows that the logarithmic rate is stable under amplitude error only with an explicit nonzero lower margin; that margin closes at a zero. The source's DQPT reading must additionally supply the physical fields of definition 10.10.

Proposition 17.5 (Local finite-N zero drift)**.** *Let $s_0 = \beta_0 + it_0$, with $0 < \beta_0 < 1$, be a zero of η of multiplicity m, and put*

$$a_m = \frac{\eta^{(m)}(s_0)}{m!} \neq 0.$$

For every fixed $C > (2|a_m|)^{-1/m}$, S_N has exactly m zeros, counted with multiplicity, in

$$|s - s_0| < CN^{-\beta_0/m}$$

for all sufficiently large N. If $m = 1$, the unique zero s_N in this disk is simple and

$$s_N - s_0 = \frac{(-1)^N (N+1)^{-s_0}}{2\eta'(s_0)} + O_{s_0}\left(N^{-2\beta_0}\log N + N^{-\beta_0 - 1}\right).$$

Proof. Choose a compact disk $U \Subset \{0 < \Re s < 1\}$ centered at s_0 and containing no other eta zero. On $|s - s_0| = CN^{-\beta_0/m}$,

$$\eta(s) = a_m (s - s_0)^m (1 + o(1)),$$

and hence

$$|\eta(s)| = (|a_m| C^m + o(1)) N^{-\beta_0}.$$

The compact-uniform Euler expansion and $N^{-\beta_0/m}\log N \to 0$ give, uniformly on the circle,

$$R_N(s) = (-1)^N \left(\frac{1}{2}(N+1)^{-s} + O_U(N^{-\Re s - 1})\right), \qquad \sup |R_N(s)| = \left(\frac{1}{2} + o(1)\right) N^{-\beta_0}.$$

The strict condition on C therefore makes $|R_N| < |\eta|$ for large N. Rouché's theorem gives exactly m zeros of $S_N = \eta - R_N$ in the disk; this is the explicit theorem 10.12 instance with $r_N = CN^{-\beta_0/m}$.

For $m = 1$, write the unique root as $s_N = s_0 + \delta_N$. The disk result gives $\delta_N = O(N^{-\beta_0})$. Expanding

$$\eta(s_N) = \eta'(s_0)\delta_N + O(\delta_N^2), \qquad (N+1)^{-s_N} = (N+1)^{-s_0}\left(1 + O(\delta_N \log N)\right)$$

in the uniform tail formula proves the stated displacement. $\square$

The eta prefactor is nonzero in the open strip, so eta and zeta multiplicities agree. The zeros of S_N are also the zeros of the normalized ideal coherence because $Z_N(\Re s) > 0$, although that normalization is not holomorphic in s. A simple critical-line zero has ideal displacement $O(N^{-1/2})$, generally in both real and imaginary parts. This does not put an exact finite-N zero on a scan constrained to $\beta_{\text{eff}} = 1/2$, furnish a calibrated five-qubit error bar, or give uniform control over zero height, multiplicity, or the strip boundary.

Corollary 17.6 (Bounded zero-set transfer). *Let Γ be a positively oriented rectifiable Jordan contour with interior D and $\overline{D} \subset \{s : 0 < \Re s < 1\}$. Suppose η has no zero on Γ, and put*

$$\sigma_\Gamma = \min_{s\in\Gamma} \Re s, \quad M_\Gamma = \max_{s\in\Gamma} |s|, \quad m_\Gamma = \min_{s\in\Gamma} |\eta(s)| > 0.$$

If

$$N^{-\sigma_\Gamma}\left(1 + \frac{M_\Gamma}{\sigma_\Gamma}\right) < m_\Gamma,$$

then S_N and ζ have the same number of zeros in D, counted with multiplicity.

Proof. Proposition 17.1 gives $|\eta - S_N| < |\eta|$ on Γ, so Rouché's theorem gives the same zero count for S_N and η, equivalently by theorem 10.11. The factor $1 - 2^{1-s}$ has no zero in $0 < \Re s < 1$, hence η and ζ have the same zeros there. □

The corollary is a positive finite-to-infinite certificate, but its premise is substantive: m_Γ must be a certified lower enclosure on the whole contour. A plot, a grid of small values, or a fitted minimum is not such an enclosure. Local-uniform convergence on compact sets also supplies no uniform large-$|t|$ claim, since the displayed modulus grows with M_Γ.

Corollary 17.7 (Exact-zero non-descent). *Let $A \subseteq \mathbb{C}$ contain 0, let $O : A \to Y$ be an observation map, and define $q(a) = \mathbf{1}_{\{0\}}(a)$. If some $z \in A \setminus \{0\}$ satisfies $O(z) = O(0)$, then there is no decision map $d : Y \to \{0,1\}$ for which $q = d \circ O$ on A.*

Proof. The fiber $O^{-1}(O(0))$ meets both the zero set and its complement, so proposition 6.4 blocks the factorization. □

Thus any positive-radius report cell containing both zero and a nonzero amplitude blocks an exact-zero decision from that report alone. This does not block an analytic identity, a certified contour count, or another independent root certificate. For stochastic reports, non-mutually-singular zero and nonzero output laws likewise block an exact decoder, and total variation gives the quantitative one-shot testing bound in proposition 6.4.

Two distinct limits. In the first construction, a DQPT is assigned through a normalized free-energy or rate-function limit after s is fixed and $N \to \infty$. The finite exponential sum in (7) is not itself the claimed thermodynamic nonanalyticity. The second construction uses the Riemann–Siegel phase $\theta(t)$ and the generalized Loschmidt amplitude

$$\mathcal{G}_N(\beta_{\text{eff}}, t) = \frac{e^{i\theta(t)} \sum_{n=1}^{N} n^{-\beta_{\text{eff}}-it} + e^{-i\theta(t)} \sum_{n=1}^{N} n^{-\beta_{\text{eff}}+it}}{2Z_N(\beta_{\text{eff}})}.$$

At $\beta_{\text{eff}} = 1/2$, its numerator is a Riemann–Siegel truncation when $N = N(t) \asymp \sqrt{t/(2\pi)}$, with the stated asymptotic remainder of order $t^{-1/4}$. This is a coupled long-time and thermodynamic regime, not the fixed-s limit above. A BSC scale certificate must separately record the integer convention for $N(t)$, normalization, convergence topology, time window, finite-size remainder, gate precision, sampling error, and order or path of limits.

Morphism chain and implementation boundary. The audit object is a chain, not one undifferentiated correspondence:

$$\text{analytic specification} \longrightarrow \text{ideal finite dynamics} \longrightarrow \text{NMR implementation}$$
$$\longrightarrow \text{scale limit} \longrightarrow \text{exact-zero or RH query.}$$

On the first edge, T prepares $\rho_{N,\beta_{\mathrm{eff}}}$ and the controlled evolution, $T^\sharp$ pairs probe σ_x, σ_y effects with the real and imaginary parts of $\mathcal{L}_N$, and K assembles the complex coherence. Equation (7) makes the ideal algebraic residual exact. The hardware edge instead owes state and pulse maps, their Heisenberg pullbacks, readout post-processing, equation and naturality residuals, the directed deficiency from the implemented to the ideal experiment, and the physical completion C. The scale edge is governed by definition 10.1; the last query is governed by definitions 6.1 and 6.3 and, for a bounded zero census, by corollary 17.6.

The reported experiment used five nuclear spins, one probe and a sixteen-dimensional four-spin working system, on a 600 MHz NMR spectrometer whose laboratory temperature was 305 K. The parameter β_{eff} in the zeta encoding was implemented by engineered level populations. It is not the inverse of the 305 K laboratory temperature; identifying the two would require an energy-unit and calibration bridge not supplied by the symbol β alone. The paper reports fitted coherence minima near five known low zeros. Its own finite-size discussion states that the coherence is nonzero and only approaches zero. This is a single published study, not an exact-zero receipt. Its data-and-code statement made material available from the corresponding author on reasonable request; no public bundle was replayed for this manuscript.

The high-zero results for $\mathcal{G}_N$, including selected zeros near the trillionth zero, are reported numerical simulations rather than executions on the NMR processor. They are neither an enumeration of the first trillion zeros nor an off-critical-line exclusion or complete zero census. Likewise, vanishing-and-revival in one observable is recurrence in the sense of return behavior. It is not BSC persistence, which requires every declared prefix identity, viability and morphism certificate in definition 9.2. Within the engineered first model, the universal statement that DQPT singularities occur only at $\beta_{\mathrm{eff}} = 1/2$ is equivalent to the RH; it remains open rather than becoming experimental output.

For BSC-ZDQ-06, the reported total bound is

$$\delta^{-1}|t|^{(1-\beta_{\mathrm{eff}})/2}\,\mathrm{Poly}\Big(\log\delta^{-1}, \log|t|, (1-\beta_{\mathrm{eff}})^{-1}, \beta_{\mathrm{eff}}^{-1}\Big)\,.$$

At $\beta_{\mathrm{eff}} = 1/2$ this gives a $t^{1/4}$ dependence against a direct Riemann–Siegel $t^{1/2}$ baseline under the paper's assumptions. It is not polynomial in the bit length $\log t$, and the article itself notes that faster classical zeta-evaluation methods exist. A quantum advantage claim therefore still requires an end-to-end, precision-matched comparison against the best applicable classical zero-location and zero-count methods.

The construction is mathematically substantive and experimentally interesting, but its logarithmic spectrum, alternating-sign observable and Riemann–Siegel phase deliberately import the arithmetic target. Realizability of that encoding does not establish an independent causal "physical origin" of the zeta zeros or a unique Kelvin temperature. Nor do finite minima or selected simulations promote the RH; BSC-ZDQ-05 blocks that inference while the status of RH itself remains open. The existing BSC morphism, quotient, scale-certificate and status machinery is sufficient for this audit; no new foundational primitive is inferred.

17.5 Four July 2026 operational-channel probes

Four contemporary studies stress different boundaries of section 14.2. They do not support one shared microscopic law. They support a common audit of preparation, driven evolution, measurement, report and decision. The fuller source crosswalk is retained in `applications/Operational_Channel_Crosswalk_2026.md`.

Table 5: Claim-local disposition of the engineered zeta–DQPT case, pinned to the article state checked 30 July 2026.

ID	Bounded claim	BSC status boundary
BSC-ZDQ-01	Exact finite identity (7)	true, proved; empirical N/A; BSC computation unexecuted; present proof and verified publication; bounded
BSC-ZDQ-02a	Alternating eta tail and local-uniform convergence	proved with the explicit half-plane and compact-set bound; no uniform large-$\|t\|$ promotion
BSC-ZDQ-02b	Bounded contour zero-count transfer	proved conditionally on a certified strict whole-boundary margin
BSC-ZDQ-02c	Fixed-s coherence exponent and pointwise rate	proved; nonuniform near zeros and not a growing-t limit
BSC-ZDQ-02d	Ambient ideal rate-singularity set and RH equivalence	proved as a representation equivalence; neither RH nor a physical DQPT is promoted
BSC-ZDQ-02d.1	Fixed-β real-time singularity slice	proved under branch visibility; no finite noisy exact-zero decision
BSC-ZDQ-02e	Multiplicity-dependent local zero drift	proved for each fixed isolated zero; no uniform height, multiplicity, or strip-boundary control
BSC-ZDQ-03	A finite-resolution report that identifies zero with a nonzero amplitude can by itself decide exact vanishing	false, proved by operational non-descent; an analytic identity or certified contour count remains possible
BSC-ZDQ-04	Five low-zero NMR coherence signatures	empirical single study; reported execution; verified publication; no BSC exact receipt or replication; local only
BSC-ZDQ-05	Finite experimental and simulated agreement entails thermodynamic exclusivity and unrestricted RH	false, proved as an inference: finite windows and selected zeros do not discharge the universal quantifier or the required limit and zero-census gates
BSC-ZDQ-06	End-to-end quantum advantage for RH verification	open and conditional on state preparation, oracle precision, sampling and the chosen classical comparator; unexecuted here; local only

Table 6: Source-local operational-channel probes.

Source	Report actually produced	Promotion not inherited
Hybrid photons [Kim et al., 2026]	Herald-conditioned, time-resolved coincidence data, measured spectral overlap and raw/model-corrected two-photon-interference visibility.	Bell-state measurement, entanglement swapping, teleportation, memory integration or scalable hybrid network.
Plasmonic PTC [Guo et al., 2026]	THz reflectivity and phase spectroscopy interpreted through a driven Floquet model, including exceptional-point mode coalescence and reported loss reduction.	Autonomous spontaneous time-crystal order, observed lasing, measured entangled plasmons or gain without pump-energy accounting.
Hiroshima alloy [Bindi et al., 2026]	Single-crystal diffraction and microprobe composition on extracted micrometre grains, including one newly assigned ordered phase.	A unique pressure–temperature–composition–time path, bulk prevalence, material properties or a manufacturing protocol.
Microwave neuron [Govind et al., 2026]	Measured broadband features, fitted token association and thresholded input-dependent stochastic bits; the image example samples a finite cache.	Lossless 8:1 coding, iid deployment bits, intrinsic semantics, cryptographic security or a live end-to-end image link.

For the hybrid-source experiment, a warm ^{133}Cs SFWM signal at 917.48 nm is matched to an InAs/GaAs quantum-dot source. The reported spectral overlap is 0.88 ± 0.01, explicitly an upper bound on full indistinguishability when other degrees of freedom are not already certified equal. The heterogeneous-source interference has raw visibility 0.41, a reported threefold rate 1.07 ± 0.14 Hz, and deconvolved visibility 0.65 ± 0.14. Heralding, an 80 ps selection window, detector response, source-rate imbalance and multiphoton contributions belong to the report channel. This is a substantive heterogeneous-interface result, not an execution of a network protocol and not evidence for BSC-QOP-03. "Without spectral modification" does not mean without tuning or selection: the emitter was thermally tuned and the report was herald-conditioned. Proposition 14.12 shows exactly why equal spectral intensities, even at overlap 1, would not certify equal photon states. The complete success/failure instrument and the normalized successful subensemble must also remain distinct.

The plasmonic experiment is an externally driven Floquet system. It reports near-unity subcycle modulation, transition through an exceptional point and more than 50% loss reduction. The source's effective-mass statement is about a model-inferred driven carrier effective-mass change, with a fitted coupling fraction and leading-order material approximation; it is not rewritten here as an unqualified vacuum-rest-mass claim. Predicted plasmonic lasing and correlated-plasmon extensions remain predictions. Proposition 14.13 retains the drive term; it does not independently replay the spectroscopy or Floquet fit, and the full bosonic cavity requires the additional truncation or domain hypotheses stated there.

The Hiroshima study inspected 34 author-supplied debris specimens and purposively extracted four roughly 8–10 μm grains from one Fe–Cr-bearing inclusion. Three were ordinary body-centred-cubic α-Fe. One was assigned an ordered AlAu4-type derivative of β-Mn (also rendered AuAl4-type in the source), space group $P2_13$, with lattice parameter 6.2666(5) Å and a measured Fe–Cr–Si–Ni–Mo–Mn–Al–P composition. These observations support the grain-local structure and chemistry, not a prevalence estimate. Formation by blast-mixed vapor condensation and rapid quenching is a source-supported historical inference; without an injective kinetic forward map, the terminal record defines an identified set rather than one certified history.

The microwave chip was fabricated in 45 nm RFSOI CMOS. The token study uses 125 physical tokens, 15,500 ordered distinct pairs and a supervised classifier on 34 features. Its association matrix is therefore conditional on the labels, split, feature construction and classifier. The probabilistic-bit study repeats all 256 8-bit patterns 300 times; a free-running oscillator supplies phase sensitivity, which does not by itself establish iid deployment bits. The image demonstration draws from a cache of those outcomes. One output bit per 8-bit pixel is nominal 8:1 lossy coding. With N draws the honest raw-bit factor is $8/N$, so $N \geq 8$ is not compression by bit count even when averaging improves PSNR.

If the scalar conditionally-iid Bernoulli model applies, then at 300 samples per input

$$\mathsf{P}\left\{\max_{x} |\widehat{q}_x - q_x| > \varepsilon\right\} \leq 512e^{-600\varepsilon^2}.$$

The simultaneous 95% radius supplied by this conservative bound is about 0.1241. Two symmetric empirical confidence intervals supplied by this bound are disjoint only when the estimated biases differ by more than 0.2482. Theorem 14.14 also blocks finite-sample zero-error recovery of all 256 symbols under that scalar model. None of this is a security theorem. If both axes of a reported association matrix denote the same tokens, proposition 14.15 additionally requires one shared permutation, not independent row and column relabelings.

The common result is therefore evidentiary, not ontological: source compatibility is observable-relative; periodic order needs a drive ledger; terminal matter does not uniquely recover history; and distortion, semantics and security are different channel claims. None of the four sources derives the fine-structure constant or gives evidential force to a numerological $1/137$ identification.

17.6 Learned models

A neural operator, dynamic mode model, or surrogate is a fitted transport with a restricted dictionary and data law. Its certificate must separate training fit, held-out prediction, full-equation residual, conservation closure, distribution shift, scale transfer, and implementation. Spectral pollution, non-identifiability, and adversarial learnability limits remain live even when visual predictions are persuasive. Conversely, an impossibility theorem for a broad task class does not demote a model with stronger verified structure.

17.7 Large proof programs as scope controls

Two contemporary cases illustrate why source status and quantifiers matter. The Wang–Zahl proof of the Kakeya set conjecture in $\mathbb{R}^3$, with a later streamlined proof, remains a preprint program at the checked date and does not settle dimensions $n \geq 4$ or all adjacent harmonic-analysis conjectures [Wang and Zahl, 2025, Guth et al., 2026]. The five-paper geometric Langlands program states a proof of the categorical global unramified conjecture in its characteristic-zero setting; the checked core papers are preprints and do not prove the arithmetic Langlands program or every local and ramified form [Gaitsgory and Raskin, 2024a, Arinkin et al., 2024a, Campbell et al., 2024, Arinkin et al., 2024b, Gaitsgory and Raskin, 2024b]. These examples are not evidence for BSC. They are models of exact scope, source labeling, and large dependency architecture.

18 Falsifier matrix

Table 7: Predeclared falsifiers and local demotion behavior.

Claim class	Decisive adverse finding	Immediate demotion	What remains
Boundary screens	failed relevance or exterior-closure screen after legal conditioning	application gate fails; sufficiency remains undecided	boundary may remain predictive, and S_{rest} still requires testing
Interior reconstruction	two inequivalent admissible interiors with identical data	identifiability refuted	forward response and quotient recovery may remain valid
Operational quotient	query differs on one exact observation class	query descent refuted	quotient valid for other descending queries
Scale transfer	actual target residual or prefix defect, or a certified lower bound, exceeds tolerance	transfer false at that scale/horizon	source and target models remain local
Persistent identity	any required prefix leaves identity or viability tube	persistence through that horizon refuted	endpoint recurrence may still hold
Koopman mode	certified full residual lower bound exceeds tolerance, or polluted eigenvalue rejected	mode claim demoted	finite predictor may still have empirical utility
Frame-change discard	two quotient-equal inputs give different transformed accessible probabilities	reduced map nonexistent on that class	paired full-system transformation remains valid
Local-to-global state	empty inverse limit or nonzero declared obstruction	global claim refuted	each local section may remain valid
Topological electric charge	two physically admissible normalizations or no derived χ	charge promotion blocked	topological invariant remains
Primordial-field QCD	failure of gauge covariance, anomaly consistency, spectra, or an independent prediction	affected physical bridge refuted	toroidal mathematics may remain
Engineered zeta–DQPT transfer	exact finite-identity mismatch under the pinned convention, certified implementation residual above budget, or a certified bounded zero-count disagreement	only the affected BSC-ZDQ node is demoted; exact-zero and RH promotion are blocked	unaffected finite algebra, limit-qualified analysis, or local empirical report remains separately typed
Operational report channel	an implemented trajectory leaves the certified reachable set, an interface changes, or a local defect/contraction bound fails	end-to-end BSC-CHN-01 certificate withdrawn	unaffected stage measurements and source-local report remain

Claim class	Decisive adverse finding	Immediate demotion	What remains
Probabilistic encoder	iid/stationarity rejected, repeated bits omitted from rate, or different same-entity row/column identities required	affected information, compression, or semantic promotion withdrawn	measured lossy hardware channel remains local
Fundamental constant from channel form	two admissible physical completions share the report envelope but carry different coupling values	derivation from ORE refuted	a separately typed electromagnetic derivation may still be attempted
Numerical verification	irreproducible build, hash mismatch, or tolerance failure	computational axis failed	mathematical statement unaffected unless execution was its only support
Mechanical proof	kernel rejection or untracked axiom/dependency	mechanical status withdrawn	informal proof may be assessed separately

19 Open problems and highest-leverage next proofs

Open problem 19.1 (Observable-complete morphism category)**.** Construct a category or double category whose objects are $(\widehat{S}_\ell, \mathscr{C}_\ell)$, whose horizontal arrows are physical interfaces, whose vertical arrows are one fixed morphism variant, and whose 2-cells certify paired observable transport. Prove closure of hard gates and identity normalization without assuming the needed coherence. In particular, resolve BSC-QOP-03 by proving or refuting closure of the full quantum variant under compatible trace-class domains, observable pullbacks, completions, and defect propagation.

Open problem 19.2 (Residual calculus for stochastic equations)**.** For nonlinear stochastic PDEs, define a weak equation-space transport $S_{\ell m}$ and residual that is stable under law-level composition. Prove an analogue of theorem 7.8 without replacing nonlinear residuals by an unjustified expectation.

Open problem 19.3 (Persistence under perturbation)**.** Find checkable conditions under which prefix operational equivalence, viability, and deficiency remain uniformly bounded over long horizons. Separate invariant laws, periodic laws, quasi-stationarity, metastability, and return recurrence.

Open problem 19.4 (Boundary reconstruction with instrument uncertainty)**.** Combine PDE stability with uncertainty in the observation kernel, finite boundary layer, calibration, and geometry. The goal is an identified set and stability modulus that accounts for all four sources without treating them as one noise scalar.

Open problem 19.5 (Operational QRF descent)**.** Characterize the state classes and accessible algebras for which a quantum reference-frame transformation factors through a declared subsystem quotient. Relate final-frame localizability to the factorization criterion of theorem 15.3.

Open problem 19.6 (Learnability-aware recurrence)**.** Join residual Koopman certification, perturbation pseudospectra, and SCI classification to produce a recurrence certificate with an honest ordered limit and finite-data stopping rule.

Open problem 19.7 (Physical observation to certified zeta-zero census)**.** For a bounded contour in $0 < \Re s < 1$, combine public raw hardware data, control programs and calibration with a uniform finite-size modulus, guarded complex root enclosures, certified boundary nonvanishing, and a complete argument-principle, Rouché, or Turing zero count. Quantify the directed deficiency from the implemented experiment to the ideal encoding and compare precision-matched end-to-end quantum and best applicable classical costs. The target is a bounded statement $\mathrm{RH}(T)$, not unrestricted RH.

Open problem 19.8 (Physical topology-to-charge bridges)**.** For any proposed carrier, derive χ from a complete action/gauge/current or measurement structure and test anomaly, boundary, quantization, and normalization conditions. A no-go result is as valuable as a construction.

Open problem 19.9 (Faithfulness of operational report envelopes)**.** Let a declared class of physical completions map to operational report envelopes under a specified intervention family. Characterize the fibers of this map, determine which additional interventions make a target property constant on each fiber, and quantify the smallest reachable-set defect that preserves the resulting decision. Treat classical, quantum and mixed measurement stages without assuming closure of the full BSC quantum morphism.

Open problem 19.10 (Non-iid physical encoders and energy-aware rate distortion)**.** For a hardware stochastic encoder with phase memory, drift and finite bandwidth, estimate the dependent joint output laws, identify minimal sufficient statistics when they exist, and derive a total-energy, total-latency and total-bit rate–distortion frontier. Compare semantic alignment only through a single identity map when both relation axes share an entity type. Keep security as a separate threat-model claim.

Open problem 19.11 (Machine-checked BSC kernel)**.** Formalize the claim-local verdict type, readiness lattices and caps, finite experiment deficiency fixture, parity-gluing obstruction, and morphism chain inequalities in a declared proof assistant. Pin the environment and retain accepted proof objects and build receipts.

The highest-leverage order is: quotient and observable descent; faithfulness of report envelopes under added interventions; morphism composition with equation-space residuals; prefix persistence; instrumented boundary stability; then domain-specific physical bridges. This order attacks hard obstructions before expensive deployment.

20 Conclusion

BSC earns its place by refusing silent transfers. Its transcribed system object describes a finite, open, instrumented domain. Its measurable repair separates raw states from equivalence, supplies context and legal filtration, and exposes the dynamics and observation kernels. The added morphism record makes state transport travel with observables, records the information destroyed by post-processing and quotient, tests target equations and commuting diagrams, requires a physical completion, and carries its own certificate.

The proved content is local but useful: quotient-respecting kernels descend; paired state and observable transport preserves accessible probabilities; statistical deficiency and naturality defects obey composition bounds; equation residuals propagate when an equation-space map exists; compatible sheaf sections glue while locally satisfiable relations may fail globally; dependency readiness caps are monotone; normalized scale rates split into normalizer and carrier exponents; continuous branch gaps expose exceptional boundaries while parameter slices may hide them; strict holomorphic contour margins transfer zero counts and local multiplicity; exact finite-label decoding requires separated observation laws; compatible fixed-interface report channels propagate implemented reachable-set defects with contraction weights; downstream channels cannot resurrect lost distinctions; driven open-system energy separates explicit drive from reduced dissipative flow; scalar Bernoulli repetition has a sufficient count and a finite-sample zero-error obstruction; one-identity relation alignment requires one shared permutation; alternating Dirichlet truncations instantiate these results with an explicit tail, pointwise exponent and rate split, ambient and fixed-line singular sets, contour count, and local root drift; compatibility reserves admit a deployment only when its frozen enclosure plus certified change remains within every claim tolerance; coupled-surrogate error inherits host sensitivity and horizon rather than standalone accuracy alone; and topology alone does not select electric charge. The fixtures make these statements vulnerable to exact tests.

Persistence is consequently neither visual recurrence nor final-time return. It is maintenance of a declared operational identity through every required fold, within viability, residual, loss, and evidence budgets. This principle may organize future theories; it has not been promoted to a universal law.

A representation may be beautiful. A mechanism must be typed. A transfer must account for its loss. A recurrence must survive perturbation. A computation must leave a receipt. A physical claim must pay the full price of its bridge.

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